Journal of Imaging

MDPI

Article

BactSegNet: A Growth-Stage-Aware, Searched Instance-Segmentation Model for *Salmonella* and *Escherichia coli* Microcolonies in Pure-Culture Bright-Field Microscopy with a Food-Matrix Surrogate

Malek Rababa [1,*], Namariq Dhahir [2], Anas AlSobeh [3], Amer AbuGhazaleh [2] and Khaled R. Ahmed [1]

[1] School of Computing, Southern Illinois University, Carbondale, IL 62901, USA; khaled.ahmed@siu.edu
[2] School of Agricultural Sciences, Southern Illinois University, Carbondale, IL 62901, USA; namariq@siu.edu (N.D.); aabugha@siu.edu (A.A.)
[3] Department of Information Systems and Technology, Utah Valley University, Orem, UT 84058, USA; anas.alsobeh@uvu.edu
* Correspondence: m.rababa@siu.edu

Abstract

Instance segmentation of small, dense, low-contrast microcolonies against confounding food matrices limits early microscopy-based screening for *Salmonella* and *E. coli*. We present BactSegNet, combining EfficientNetV2, BiFPN, selective-kernel attention, and growth-stage FiLM conditioning based on a learnable embedding of recorded acquisition stage. BE-NAS explored a 2304-configuration search space and evaluated 71 unique architectures. Across twelve retrained candidates, proxy and full-training rankings agreed (Spearman $\rho = 0.867$ for mAP@50 and $\rho = 0.790$ for mAP@50–95). Across the 71 architectures evaluated in the BE-NAS proxy search, three-seed proxy fitness showed high reliability ($\mathrm{ICC}(1,1) = 0.91$; 213 seed-level evaluations), while phenotype caching reduced 180 scheduled evaluations to 71 unique trainings (213 rather than 540 seed-level proxy runs). On 14,318 expert-verified polygon instances spanning six stages, two species, and onion-present/absent conditions, BactSegNet-L achieved mask mAP@50 = 0.9636 and mAP@50–95 = 0.7154, 3.56 percentage points above a multi-seed YOLOv8 baseline and higher mAP@50 than sixteen class-aware single-run references across three paradigms. A capacity-preserving stage-perturbation control reduced mAP@50 from 0.9636 ± 0.0049 with correct stage assignments to 0.9487 ± 0.0040 after shuffling image–stage correspondence across three seeds. Gains reached 8.66 pp for *Salmonella* at 1.5 h and 4.90 pp on onion. Across 2756 matched instances, area MAPE was 12.15% with $r = 0.997$, dominated by the largest instances. BactSegNet-Efficient retained 0.9510 mAP@50 with 19.1% fewer FLOPs at 43.9 fps. The main dataset comprises pure-culture frames; an independent, expert-annotated mixed-culture evaluation (400 images, 3.5 and 4.0 h) yielded mask mAP@50=0.8857 ± 0.0206 across three seeds without retraining. Recorded growth-stage metadata are required at inference, unlike the image-only comparison models. These results support early, physically calibrated microcolony segmentation while distinguishing the original pure-culture evaluation from the additional mixed-culture test.

Keywords: deep learning; instance segmentation; foodborne pathogen detection; bacterial microscopy; neural architecture search; food safety; microcolony segmentation; *Salmonella*; *Escherichia coli*; computer vision

Check for updates

Academic Editor: Jérôme Gilles

Received: 31 August 2026
Revised: 15 October 2026
Accepted: 21 October 2026
Published: 23 October 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Foodborne bacterial contamination remains a major public-health burden. The World Health Organization attributes 600 million foodborne illnesses and 420,000 deaths annually to contaminated food, with non-typhoidal *Salmonella enterica* responsible for the largest share of foodborne deaths [1]. *Salmonella* and Shiga toxin-producing *E. coli* (STEC) are also recurrent causes of multi-state outbreaks associated with fresh produce [2,3]. Conventional culture-based enumeration is accurate but typically requires 24–48 h before colonies can be counted and a positive confirmed [4], motivating automated microscopy methods that can detect microcolonies earlier.

The image-analysis problem is difficult for three reasons. First, object scale changes strongly with growth: in our data, the median annotated area increases 38.7-fold from 1.5 to 4.0 h, with different trajectories for the two organisms. Second, onion-tissue edges and granular structures can resemble the targets, creating a visually confounding background. Third, appearance changes systematically with incubation time, but conventional segmentation pipelines leave this growth context implicit in the same visual feature stream rather than estimating it explicitly and using it to modulate the segmentation representation. These difficulties also arise in other scientific-imaging tasks involving small, dense objects and strong scale variation.

Bright-field microscopy is inexpensive and widely available, but early screening requires more than object detection. Pixel-level masks are needed to separate nearby microcolonies and to measure their size, whereas bounding boxes cannot directly support colony-area monitoring or growth quantification. Instance segmentation is therefore the appropriate formulation because it combines localisation, species assignment, and mask prediction under low contrast and background clutter.

Deep learning provides several candidate paradigms. Mask R-CNN [5] offers accurate two-stage segmentation; YOLO-based models [6,7] provide a strong real-time accuracy–speed trade-off; detection transformers [8,9] offer anchor-free alternatives; and foundation models such as SAM and Cellpose-SAM [10,11] provide general-purpose segmentation. Biomedical adaptations have improved feature fusion, attention, crowded-object handling, and bacteria-specific segmentation [12–15], while food-relevant studies increasingly apply deep models to bacterial microscopy [16–18].

Three gaps motivate this work. First, architecture components are usually chosen from natural-image defaults rather than configured jointly for the object-scale statistics of a specialised microscopy task; when low-fidelity neural architecture search (NAS) is used, its ranking proxy is often assumed rather than validated on the target problem [19,20]. Second, bacterial segmentation models generally leave growth-stage information implicit in the main visual representation rather than extracting a dedicated representation of growth-related visual context and feeding it back into the segmentation hierarchy. Third, aggregate evaluation can hide the conditions that determine practical value, especially the earliest growth stages, the target species, the food-matrix background, and whether predicted masks recover physically meaningful object size.

We address these gaps with BactSegNet, a one-stage instance-segmentation model for *Salmonella* and *E. coli* microcolonies under the pure-culture acquisition protocol used here. The model combines an EfficientNetV2 backbone [21], BiFPN [22], selective-kernel feature-pyramid attention (SK-FPA) [23], and a growth-stage-aware attention (GSAA) module that maps the recorded acquisition stage to a learnable embedding and applies FiLM-style conditioning [24] to the multi-scale features on a YOLOv8 segmentation head [6]. The configuration is selected by BE-NAS, a budget-efficient evolutionary search over backbone capacity, neck width and depth, attention branching, and conditioning capacity. Its adaptive proxy-epoch evaluation (APED) is validated against full training rather than treated as an unverified assumption. For GSAA conditioning, the recorded acquisition stage is supplied

as metadata at training and inference and mapped to a learnable stage embedding; image pixels remain the sole visual input.

Evaluation is deliberately stratified by growth stage, species, and onion-present/absent background condition. Because the imaging is physically calibrated, we additionally test whether predicted masks recover object area in μm^2. One scope condition applies throughout: every frame is a pure culture of a single organism acquired under either onion-present or onion-absent background conditions. The results therefore establish detection, segmentation, and species assignment under this protocol, not discrimination of both organisms within a single co-contaminated field.

The raw bright-field imagery is drawn from an evolving microscopy corpus used in prior studies with different task formulations [16–18]; we therefore do not claim novelty of the raw images. The present study instead uses a re-curated, expert-verified CVAT polygon annotation set covering all six growth stages and both background conditions, together with a leakage-audited partition and stratified instance-segmentation evaluation.

The main contributions are:

1. **A searched, scale-adaptive, growth-context-conditioned architecture.** Jointly configuring EfficientNetV2, BiFPN, SK-FPA, and GSAA yields mask mAP@50 = 0.9636, 3.56 pp above the multi-seed YOLOv8 baseline. A comparable-capacity analysis and a cost-aware re-ranking show that the gain is not explained by model size alone; BactSegNet-Efficient retains 0.9510 mAP@50 with 19.1% fewer FLOPs.
2. **A validated low-fidelity search proxy.** Across twelve fully retrained candidates, proxy ranking agrees with full training at Spearman $\rho = 0.87$. Across the 71 proxy-evaluated architectures, with three seed-specific proxy-fitness measurements per architecture (213 evaluations), $\mathrm{ICC}(1,1) = 0.91$, and retrospective APED analysis identifies epoch 4 as a sufficient proxy budget for subsequent searches, while the reported search itself used the full ten-epoch trajectories.
3. **Budget-efficient evolutionary search with phenotype caching.** Over a 2304-configuration space, 180 scheduled evaluations were served by 71 unique architecture trainings (213 rather than 540 seed-level proxy runs) at a total search cost of 53.8 GPU-hours. The selected configuration differs from the expert-seeded designs on three of seven genes.
4. **Growth-stage-aware conditioning.** GSAA maps the recorded acquisition stage to a learnable embedding and uses it for FiLM-style modulation. Relative to the image-only YOLOv8s baseline (0.9280), BactSegNet without GSAA reaches 0.9445 (+1.65 pp), and full GSAA reaches 0.9636 (a further +1.91 pp). A simple stage-conditioning baseline reaches 0.9590, indicating that most of the conditioning benefit comes from the stage information rather than the particular FiLM formulation. Removing GSAA reduces mAP@50 by 1.91 pp, with a larger effect for *Salmonella* (−2.76 pp) than *E. coli* (−1.07 pp). In a capacity-preserving control, shuffling image–stage correspondence reduces mAP@50 from 0.9636 ± 0.0049 to 0.9487 ± 0.0040 across three seeds (−1.49 pp), supporting a contribution from stage-specific conditioning rather than added capacity alone.
5. **An expert-verified microscopy dataset with stratified evaluation.** The 14,318-instance annotation set supports analysis by stage, species, and background. The gain reaches +8.66 pp for *Salmonella* at 1.5 h and +4.90 pp for *Salmonella* on onion, while the models are essentially tied for *Salmonella* at 3.5 h (−0.16 pp), showing where the advantage does and does not occur.
6. **Physically calibrated area outputs.** Mask area is computed deterministically from the calibrated pixel grid, with no learned area head. Across 2756 matched instances,

MAPE is 12.15% and $r = 0.997$; per-stage MAPE decreases from 17.19% at 1.5 h to 3.35% at 4.0 h. The inference output is therefore $\{\text{species}, \text{mask}, \text{area}_{\mu m^2}\}$.

The remainder of this paper reviews related work (Section 2), describes the dataset, architecture, search, and evaluation protocol (Section 3), reports the experiments (Section 4), and discusses their interpretation, limitations, and conclusions (Sections 5 and 6).

2. Related Work

2.1. Deep Learning for Bacterial and Cellular Segmentation

Automated analysis of microorganisms in microscopy has a long history, and deep learning has substantially improved its accuracy and throughput. Early convolutional approaches addressed colony-level tasks such as counting and segmentation: U-Net-based models have been used for robust automated colony-forming-unit counting [25], and dedicated colony-detection datasets and detectors have since been released to standardise the problem [26]. More recently, one- and two-stage detectors have been applied directly to bacterial imagery—for example, YOLO-family models for rapid colony counting [27], and Mask R-CNN and its variants for cell- and nucleus-level instance segmentation [5]. A parallel line of work targets food safety specifically, classifying foodborne bacteria at the microcolony stage to shorten the time between sampling and a result [28,29] and estimating bacterial growth from time-lapse microscopy [30].

Within this food-safety microscopy setting, prior work on the same evolving *Salmonella*/*E. coli* corpus provides the closest context. Because these studies share corpus provenance, their relationship to the present work requires explicit distinction. Koirala et al. [16] applied Mask R-CNN under an earlier segmentation-annotation regime that combined manual and semi-automated procedures with manual verification; that study also did not use *Salmonella* on the onion background at 1.5 and 2.0 h for model training because those early colonies were difficult to distinguish from onion particles. Rababa et al. [17] used the broader corpus for a bounding-box detection study of backbone and feature-fusion design rather than polygon instance segmentation, while AlSobeh et al. [18] addressed classification of co-contaminated samples. The present work does not claim novelty of the raw microscopy corpus and does not count quantitative results from those publications as results of this study. Instead, it uses the re-curated, expert-verified CVAT polygon annotations described here, retains all six growth stages and both background conditions, and evaluates the models under the present image-level partition and stratified instance-segmentation protocol. Thus, any overlap is in the provenance of the evolving raw-image corpus; the task formulation, retained polygon annotation set, evaluation protocol, and quantitative results reported here are those of the present study.

A distinct and highly relevant strand of this literature is concerned less with detection than with the geometry of crowded biological objects, and it is instructive because it isolates the same difficulty that our task presents. StarDist parameterises each object as a star-convex polygon, which allows touching and overlapping nuclei to be separated where pixel-grouping methods merge them [13]. Cellpose learns a flow field toward cell centres and thereby generalises across cell shapes and imaging modalities [31], and Omnipose extends this idea specifically to bacteria, using distance-field gradients to segment elongated, branched, and antibiotic-perturbed morphologies on which its predecessor fails [14]. MiSiC targets the complementary problem of high-throughput segmentation of dense bacterial communities [15]. These methods establish that bacterial morphology is diverse enough to warrant dedicated shape representations and that dense packing—rather than raw image quality—is often the binding constraint. They also share two properties that matter here: they are trained and evaluated predominantly on stained, phase-contrast, or fluores-

cence preparations with high object–background separability, and they are class-agnostic, producing masks without a species label.

Two limitations therefore recur across this literature when it is viewed from the perspective of food-safety screening. First, most methods operate on clean or stained preparations—Gram staining, phase contrast, or agar plates—rather than on unstained cells imaged directly against a real food matrix, where background structures mimic the target organisms and there is no explicit supervision for the distractor class. Second, the majority of studies treat the problem as detection, counting, classification, or class-agnostic segmentation, whereas food-safety quantification requires the pixel-level separation of dense, overlapping, morphologically similar cells together with a species assignment. Instance segmentation of *Salmonella* and *E. coli* against a produce background, across the early growth stages that matter most for screening, remains comparatively unexplored.

2.2. YOLO Instance Segmentation with Attention and Feature Fusion

Among one-stage frameworks, the YOLO family has become the dominant choice for real-time instance segmentation because of its favourable speed–accuracy trade-off [6,7]. A large body of work improves these models for small, dense objects by strengthening two components in particular: the multi-scale neck and the attention mechanism. On the neck side, feature-pyramid designs such as FPN [32], PANet [33], and the search-derived NAS-FPN [34] progressively improve how features at different resolutions are fused. On the attention side, channel and spatial modules such as SENet [35], CBAM [36], and selective-kernel networks [23] allow a network to emphasise informative channels and adapt its effective receptive field. Recent cell-segmentation models combine these ideas: ASF-YOLO, for instance, augments a YOLO segmentation framework with a scale-sequence feature-fusion module and a channel-and-position attention mechanism, achieving strong results on cell instance segmentation [12]. Similar attention-enhanced YOLO variants have been applied to microorganism detection in complex microscopy backgrounds [37].

These works establish the components on which we build—an EfficientNetV2 backbone [21], a BiFPN neck [22], selective-kernel attention, and a YOLOv8 segmentation head. Two limitations motivate our departure from them. First, they configure these components by hand and validate them on natural-image or general cell benchmarks; the receptive-field behaviour, neck width and depth, and attention capacity that are appropriate for microcolonies whose area spans nearly two orders of magnitude within one dataset are not obvious a priori and are not obtained by transferring a fixed configuration from another domain. Second, these models generally use a single visual pathway in which growth-related context must remain implicit in the same features used for localisation and mask prediction. We instead make growth context explicit by mapping the recorded acquisition stage to a learnable embedding and use feature-wise linear modulation (FiLM) [24] to inject that stage representation into the multi-scale segmentation features.

2.3. Transformer Detectors and Segmentation Foundation Models

Two further paradigms have emerged alongside the CNN detectors above, and both are represented in our comparison. The first is the detection transformer, introduced by DETR [8], which removes hand-crafted anchors and non-maximum suppression by casting detection as direct set prediction. Subsequent work has closed the gap in convergence speed and real-time throughput that limited early DETRs; RF-DETR [9] is a recent example that pairs a self-supervised DINOv2 backbone [38] with weight-sharing architecture search to select accuracy–latency operating points for a target dataset and provides an instance-segmentation variant. It is a useful comparator here for two reasons: it is a strong, current model on a different architectural paradigm, and, like our own approach, its configuration

is obtained by search rather than by hand—so the comparison is between two searched designs rather than between a searched model and hand-designed ones.

The second paradigm is the segmentation foundation model. The Segment Anything Model (SAM) [10] demonstrated promptable, class-agnostic segmentation generalising across natural-image domains, and the microscopy community has since adapted this class of model to cellular data. Cellpose established a generalist flow-based framework for cellular segmentation [31], and Cellpose-SAM [11] replaces its backbone with SAM's pretrained transformer, reporting generalisation that exceeds inter-annotator agreement on standard cellular benchmarks. Whether such a generalist transfers to unstained bacteria imaged against a food matrix is an open question with direct relevance to domain transfer, and it is one we examine empirically in Section 4.1. Note that Cellpose-SAM is class-agnostic and does not assign a species label, so it addresses only the segmentation half of the problem posed here. The broader question this comparison speaks to—whether large-scale generalist pretraining removes the need for task-specific architecture design in specialised, low-contrast, domain-shifted imaging—is one of current interest well beyond bacteriology, and our result provides one carefully controlled data point on it.

2.4. Neural Architecture Search and Low-Fidelity Proxy Evaluation

Neural architecture search (NAS) automates the design of network components that would otherwise be tuned by hand [19,39]. Evolutionary approaches are a well-established branch of NAS: regularised evolution and large-scale evolutionary search demonstrate that population-based mutation and selection can discover competitive image classifiers [40,41]. Because evaluating every candidate by full training is prohibitively expensive, NAS methods rely on low-fidelity proxies—training briefly, at reduced resolution, or on a subset of data—and on early-stopping schemes such as successive halving and Hyperband that allocate computation preferentially to promising candidates [42,43]. A parallel line of work pushes this further with zero-cost proxies that score an architecture from a single minibatch [20]. Across both lines, the quantity that determines whether a proxy is usable is the rank correlation between proxy and full-training performance, and the literature is explicit that this correlation is task- and search-space-dependent and can degrade sharply when a proxy is transferred to a new setting [19,44]. In practice, however, a proxy is frequently adopted for a new application domain without being re-measured there.

Building on this foundation, we design and apply a task-specific evolutionary search, BE-NAS, that jointly configures the backbone, neck, attention, and conditioning of our segmentation model. Our contribution is not a new search paradigm, nor a comparison of search paradigms, but the careful adaptation of an established one to this problem, together with the evidence needed to trust it: an adaptive proxy-epoch evaluation (APED) criterion that specifies when a candidate's proxy run may terminate early, an in-loop phenotype-hash cache that avoids retraining recurring architectures, and—most importantly—an empirical validation on this task that the resulting proxy ranking agrees closely with full-training performance and separates architectures well above the level of seed noise. To our knowledge, architecture search has not previously been applied to, nor its proxy validated for, instance segmentation of bacteria in a food matrix.

3. Materials and Methods

3.1. Dataset

3.1.1. Image Acquisition and Composition

The images used here are drawn from an evolving bright-field microscopy corpus that has supported previous studies with different task formulations and annotation regimes [16–18]; the raw imagery is therefore not presented as newly acquired for this study.

For the present work, the corpus was re-curated as an expert-verified instance-segmentation dataset of *Salmonella* and *Escherichia coli* imaged against onion-tissue backgrounds, reflecting the conditions of early food-safety screening. Images were acquired on an Olympus IX73 inverted microscope (Olympus Corporation, Tokyo, Japan) with a 60× objective and a colour camera and stored as RGB TIFFs of 2160 × 1620 pixels. At this magnification, the pixel pitch is 0.125 µm, so one pixel corresponds to 0.015625 μm^2 and a full frame covers a field of view of approximately 270 × 202.5 μm^2. This calibration lets us report object sizes in physical units (μm^2) rather than pixels, which is what makes the scale analysis of Section 3.1.3 interpretable and comparable across imaging setups.

Each sample was imaged as a pure culture of one organism, and every frame is additionally characterised by two acquisition variables used to describe and stratify the dataset: the incubation time, sampled at six growth stages $s \in \{1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ hours, and the background condition, either onion-present (cells imaged over onion tissue) or onion-absent (cells imaged over a clean medium). The recorded incubation time is mapped to one of the six acquisition stages and supplied to GSAA as conditioning metadata at training and inference (Section 3.2.5). The background condition defines the clutter regime against which false positives are most likely and is treated as an explicit axis of evaluation. Sampling both organisms across all six stages and both backgrounds yields a controlled grid in which object scale, morphology, and background clutter vary systematically rather than incidentally.

The dataset was collected across multiple biological samples and replicate plates. Specifically, 20 onion samples were used, with three replicate plates prepared per sample, and experiments were conducted across three independent days. The six growth stages analysed in this study were obtained from separate experimental plates rather than by repeatedly imaging the same plate or microscopic field over time. For each growth stage, between 90 and 100 images were collected for each bacterial species under each background condition (onion-present and onion-absent).

Microbiological Preparation

The microbiological preparation followed the published acquisition protocol for this microscopy corpus [45]. Two Gram-negative strains were used: *Salmonella enterica* serovar Typhimurium and non-pathogenic *Escherichia coli* K-12. Frozen stocks were streaked onto tryptic soy agar (TSA) and incubated at 37 °C for 18–24 h. A single colony was then inoculated into 10 mL tryptic soy broth (TSB) and incubated overnight at 37 °C with shaking at 175 rpm. Overnight cultures were diluted with sterile Dulbecco's phosphate-buffered saline (DPBS) to approximately 10^6 CFU/mL, as confirmed by optical-density calibration. Soft Salmonella–Shigella agar and soft HiCrome™ Chromogenic Coliform Agar were used for *S.* Typhimurium and *E. coli* K-12, respectively. Approximately 2.5 µL of bacterial culture was deposited with or without onion tissue, and inoculated plates were incubated at 37 °C for up to 4 h. The use of known bacterial strains together with the corresponding selective media provided species-specific control of the cultures used for imaging.

3.1.2. Annotation and Expert Verification Protocol

The dataset is annotated for instance segmentation: every bacterial object is delineated by a closed polygon that follows its boundary, and each polygon carries one of two class labels, 0 for *Salmonella* and 1 for *E. coli*. Polygon delineation was carried out in CVAT, and the resulting polygons were exported directly as the mask ground truth; from each polygon, we also derive an axis-aligned bounding box, so box- and mask-based analyses share a single, consistent ground truth. This pipeline from object to polygon, derived box, and class label is illustrated in Figure 1. Deriving both supervision signals from one delineation

removes a common source of inconsistency between box- and mask-based supervision and evaluation. Compared with the earlier segmentation study, which combined manual freehand annotation with Cellpose- and Mask R-CNN-assisted annotation followed by manual verification [16], the retained polygons in the present work follow a dedicated CVAT-based, expert-supervised annotation and verification protocol. The earlier study also excluded *Salmonella* on the onion background at 1.5 and 2.0 h from model training because those conditions were difficult to annotate; those early-stage conditions are annotated, retained, and evaluated here.

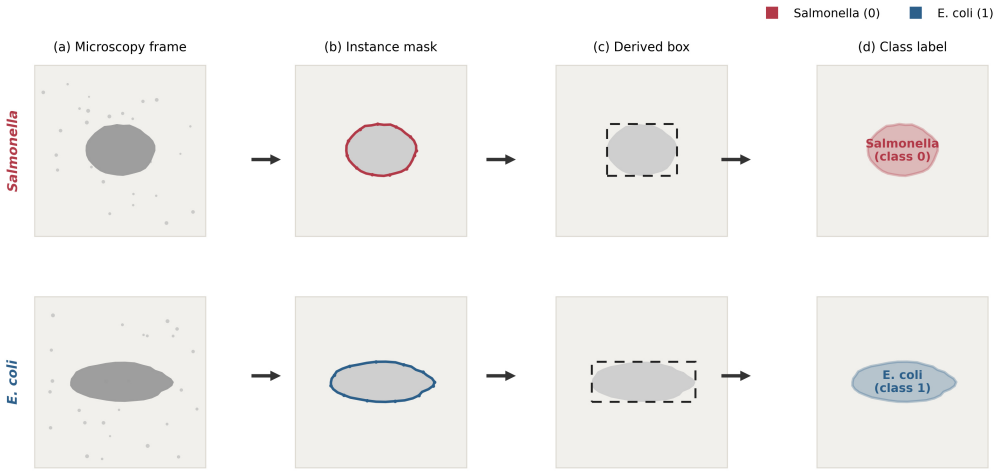


Figure 1. Annotation protocol. Each annotated object in a microscopy frame (**a**) is delineated by a closed instance-segmentation polygon (**b**); an axis-aligned bounding box is derived from the polygon (**c**); and each instance is assigned one of two class labels, 0 (*Salmonella*) or 1 (*E. coli*) (**d**). Deriving both mask and box ground truth from a single delineation keeps the two supervision signals mutually consistent. The protocol is identical for both organisms and is shown for each: *Salmonella* ((**top row**), class 0) and *E. coli* ((**bottom row**), class 1), whose compact and elongated microcolony morphologies are delineated under the same rules.

Annotation quality is the ceiling on any supervised segmentation result, and it is a particularly binding constraint in this setting: at the earliest incubation times, a microcolony and an onion-tissue particle can be similar in size, brightness, and edge profile, so labels produced without domain input can encode the annotator's uncertainty as if it were data. We therefore treated verification as part of the protocol rather than as a final check. All polygon annotations were produced and then reviewed and cross-checked under the supervision of a domain expert in the biological sciences, and only frames whose labels passed this verification are used in the experiments reported here. Frames that could not be labelled to that standard were withheld rather than labelled speculatively. This verification was a quality-control review rather than a formal independent inter-annotator agreement study, and no separate agreement statistic is claimed. The practical consequence is a reduced risk that apparent model errors are instead consequences of uncertain reference labels, which is important for the stratified, per-condition analysis in Section 4: a per-stage comparison is only informative if annotation reliability does not itself vary with stage.

Because imaging begins after 1.5 h of incubation, the annotated objects are microcolonies rather than individual cells: the median annotated object already spans 12.5 µm at the earliest sampled stage, and only 5 of the 14,318 annotated objects (0.03%) have a longest dimension below 3 µm. We therefore use object and microcolony interchangeably in what follows and reserve cell for the individual organisms that compose them.

3.1.3. Scale and Characteristics of the Dataset

Two properties of the data motivate the architectural choices of Section 3.2, and both are quantified in Table 1.

Table 1. Scale of the annotated objects by growth stage, computed over all 14,318 expert-verified instances. Areas are polygon areas in μm²; IQR is the interquartile range, i.e., the interval between the 25th and 75th percentiles of the area distribution within each group; "longest dimension" is the larger of the polygon's width and height. The two organisms diverge sharply after 2.0 h: over the full window, the median *E. coli* object grows 87.5-fold against 13.8-fold for *Salmonella*. Fold changes are computed from unrounded medians and may differ slightly from ratios formed with the rounded values shown.

Stage	*n*	Area (μm²)		Longest Dim.	Median Area (μm²)	
		Median	IQR		*Salmonella*	*E. coli*
1.5 h	4212	51	36–73	12.5	54	50
2.0 h	4312	66	41–116	13.5	47	111
2.5 h	2011	212	105–384	25.1	110	374
3.0 h	1502	125	72–651	18.0	91	1302
3.5 h	1160	871	429–3089	43.0	534	3317
4.0 h	1121	1972	526–4485	60.8	740	4340
All	14,318	84	48–279	16.5	70	129

First, the annotated objects are small and low-contrast relative to the field of view: the median instance has an area of 84 μm² (about 5400 pixels) and a median longest dimension of 16.5 μm, against a 270 × 202.5 μm² frame, with an interquartile range (IQR, the interval between the 25th and 75th percentiles) of 48–279 μm² in area. The multi-scale neck and the adaptive receptive field of the SK-FPA module (Section 3.2.4) are directly targeted at this dominant object scale.

Second, object size grows markedly over the incubation window, though not monotonically at every stage: the median instance area rises from 51 μm² at 1.5 h to 1972 μm² at 4.0 h, a factor of 38.7, with a local dip at 3.0 h (Table 1). Critically, the two organisms do not follow the same trajectory. Over the same window, the median *E. coli* instance grows 87.5-fold (50 → 4340 μm²) while the median *Salmonella* instance grows only 13.8-fold (54 → 740 μm²), so two objects imaged at the same incubation time can differ in area by nearly an order of magnitude depending on species. A single receptive field cannot be well matched to both regimes at once, and the growth stage alone does not determine the expected object scale without knowing the species. This growth-driven morphological drift, and the species-dependence of its rate, is precisely the variable the GSAA module conditions on (Section 3.2.5), and it is why early-stage frames are simultaneously the most important for screening and the hardest to segment.

3.1.4. Data Partitioning and Training Protocol

We partition the verified frames into training, validation, and test splits in a 70:10:20 ratio at the image level. To audit identifier-level partition leakage, we extracted the filename stem of every image in the training, validation, and test directories and compared the resulting identifier sets pairwise. All three intersections were empty, and the number of unique image identifiers equalled the sum across the three partitions, confirming that no image identifier appeared in more than one split. This audit verifies identifier-level partition separation; it does not constitute a content-hash or pixel-level duplicate-image audit. Images from the same biological sample, plate, or acquisition session may occur in different partitions, as sample-, plate-, and session-level separation cannot be retrospectively verified. The proxy runs that drive the BE-NAS search (Section 3.2.7) and the final training

of BactSegNet both optimise on the training split and select on the validation split, and every reported test metric is computed once on the held-out test split. All models are trained at an input resolution of 1024 × 1024 under a single optimisation recipe, summarised in Table 2, so that differences in performance reflect the architecture rather than the training schedule. To account for optimisation variance, the baseline, the three BactSegNet variants, and each ablated configuration are trained with three independent random seeds, and we report the mean over seeds.

Table 2. Training and evaluation configuration. Unless otherwise stated, the same training and evaluation settings were applied to all models reported in this study. Values marked with † are unmodified Ultralytics defaults. The evaluation confidence threshold of 0.001 is the validation threshold used to construct the precision–recall curve for average-precision computation, not an operating threshold for deployment. The backbone-initialisation entry applies to BactSegNet only: the YOLO-family models were initialised from their respective released pretrained checkpoints, so initialisation is matched within each family but not across families. For BactSegNet, the growth stage is supplied to GSAA from the acquisition metadata. Mosaic sampling is restricted to source images from the same stage so that each composite has a valid single stage assignment.

Group	Setting	Value
Schedule	Input resolution	1024 × 1024
	Epochs	100
	Batch size	8
	Early-stopping patience	50 epochs
	Data-loader workers	8
	Random seeds	three per configuration (Section 3.3)
	Backbone initialisation	BactSegNet: ImageNet-21k pretrained, fine-tuned on ImageNet-1k (see caption)
Optimisation	Optimiser	AdamW
	Initial learning rate η_0	1×10^{-3}
	LR scheduler	linear decay †
	Final LR factor	0.01 (i.e., $\eta_0 \rightarrow 1 \times 10^{-5}$)
	Momentum (β_1)	0.937
	Weight decay	5×10^{-4}
	Warm-up	3 epochs †
Precision & masks	Mixed precision (AMP)	enabled
	Deterministic mode	enabled
	Overlapping masks	enabled
	Mask ratio	4
Augmentation †	HSV (h, s, v)	0.015, 0.7, 0.4
	Translate/scale	0.1/0.5
	Horizontal flip	0.5
	Vertical flip, rotation, shear	disabled
	Mosaic/mixup/copy-paste	1.0/0/0
	Mosaic disabled for final	10 epochs
Evaluation †	Confidence threshold	0.001
	NMS IoU threshold	0.7
	Maximum detections	300 per image

Two aspects of the recipe are worth making explicit. First, no hyperparameter was tuned per architecture: the batch size, learning-rate schedule, weight decay, and augmentation policy are identical for every model in every comparison, including the thirteen YOLO-family baselines, so no model benefits from a more favourable configuration. Second, the augmentation and non-maximum-suppression settings are the framework defaults rather than values tuned on this dataset; we report them in Table 2 for reproducibility and note that leaving them untouched removes a further degree of freedom from the comparison.

The held-out test split contains 478 images and is close to balanced across every axis of interest—237 *Salmonella* and 241 *E. coli* images, 237 onion-absent and 241 onion-present—with between 77 and 81 images at each of the six growth stages so that stage-, species-, and background-stratified metrics are each computed on comparable sample sizes. The dataset comprises 14,318 expert-verified instances (7493 *Salmonella* and 6825 *E. coli*); their joint distribution over the growth stage, species, and background condition is shown in Figure 2, and the per-split image counts are summarised in Table 3; representative frames with their ground-truth segmentation overlays, for the two organisms at three representative growth stages, are shown in Figure 3.

Table 3. Composition of the expert-verified segmentation dataset used to train and evaluate BactSeg-Net. Splits follow a 70:10:20 image-level partition. The test split is close to balanced across species, background condition, and growth stage.

	Train	Val	Test
Images (total)	1673	239	478
Test-split composition			
Salmonella images	–	–	237
E. coli images	–	–	241
Onion-absent images	–	–	237
Onion-present images	–	–	241
Images per growth stage	–	–	77–81

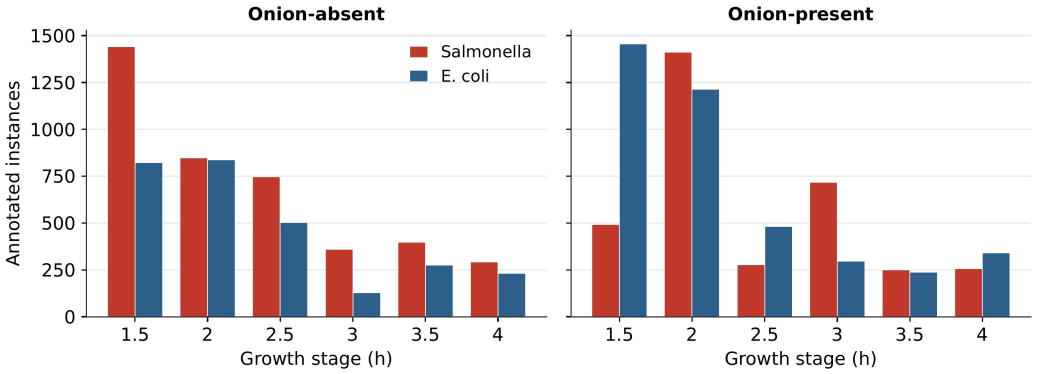


Figure 2. Distribution of the 14,318 expert-verified ground-truth instances over the growth stage, species, and background condition. Annotated objects are most numerous at the early growth stages, where microcolonies are dense and small, and become fewer as they enlarge at later stages; the two organisms are annotated in comparable numbers overall (7493 *Salmonella* and 6825 *E. coli*).

The stage-dependent background proportions shown in Figure 2 represent instance-level distributions rather than image-level sampling proportions. Image acquisition was approximately balanced across the onion-present and onion-absent conditions, whereas the number of annotated microcolony instances varied across images and growth stages.

Independent Mixed-Culture Evaluation Set

In addition to the original pure-culture dataset, we evaluated the trained models without retraining on a separate set of 400 expert-annotated mixed-culture images containing 1509 annotated instances. Both bacterial species occur within the same fields of view. The set covers 3.5 and 4.0 h and onion-present and onion-absent conditions and was obtained from independent biological samples or plates. It was not used for training, validation, or model selection. The evaluation is restricted to these two stages, for which the reference annotations were suitable for evaluation.

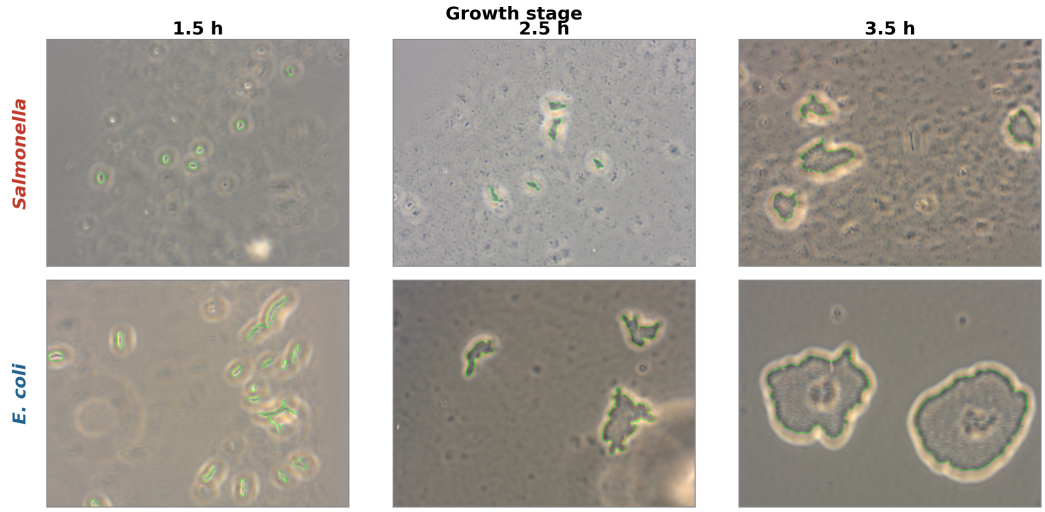


Figure 3. Representative frames from the dataset with their ground-truth instance-segmentation overlays for the two organisms (*Salmonella*, **top** row; *E. coli*, **bottom** row) at three of the six sampled growth stages (1.5, 2.5, and 3.5 h), chosen to span the range of object scale in the dataset. Mask colour denotes the class label.

3.2. The BactSegNet Architecture

3.2.1. Overall Architecture

BactSegNet is a one-stage instance-segmentation model built on the YOLOv8 segmentation framework. Like other YOLO segmentation networks, it consists of three parts—a backbone, a neck, and a segmentation head. The backbone extracts features at five successive strides, denoted P1–P5; only the three deepest levels (P3, P4, P5, at strides 8, 16, and 32) are forwarded to the neck, which fuses them across scales; the head then predicts, at each scale, class-wise bounding boxes together with prototype-based instance masks. This generic layout fixes the vocabulary and dataflow that the proposed model builds upon.

The complete proposed architecture is shown in Figure 4. Given a bright-field input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, an EfficientNetV2-L backbone produces three multi-scale feature maps $\{\mathbf{C}_3, \mathbf{C}_4, \mathbf{C}_5\}$ at strides 8, 16, and 32. Because these levels have heterogeneous channel widths, each is first projected to a common width C by a 1×1 convolution, after which a BiFPN neck fuses them through combined top-down and bottom-up pathways. Each fused level is then refined by a Selective-Kernel Feature-Pyramid Attention (SK-FPA) module and modulated by a Growth-Stage-Aware Attention (GSAA) module conditioned on a learnable embedding of the recorded acquisition stage. Finally, the YOLOv8 segmentation head combines a set of prototype masks generated from the highest-resolution level with per-instance mask coefficients predicted at all three levels, producing the final instance masks for *Salmonella* and *E. coli*. The specific configuration of these components is not fixed by hand but selected by the BE-NAS search of Section 3.2.7. The following subsections describe each component in turn.

3.2.2. Backbone

The backbone is EfficientNetV2-L [21], used as a feature extractor with its ImageNet-21k pretrained weights. EfficientNetV2 combines fused and standard mobile inverted bottleneck (MBConv) blocks under a compound scaling rule that balances depth, width, and resolution, giving a strong accuracy-to-cost ratio and stable training on limited data. We extract the feature maps at strides 8, 16, and 32, which, for a 1024×1024 input, correspond to spatial resolutions of 128×128, 64×64, and 32×32 with 96, 224, and 640 channels, respectively (levels P3, P4, P5). The two shallowest levels, P1 and P2, are computed as part

of the backbone but are not forwarded to the neck. Their high spatial resolution makes them comparatively costly; the reported architecture therefore uses P3–P5 as the three feature-pyramid inputs, without making a separate causal claim about the independent contribution of P1 or P2. The choice of EfficientNetV2-L over the smaller -S and -M variants was made by the architecture search rather than fixed in advance (Section 3.2.7).

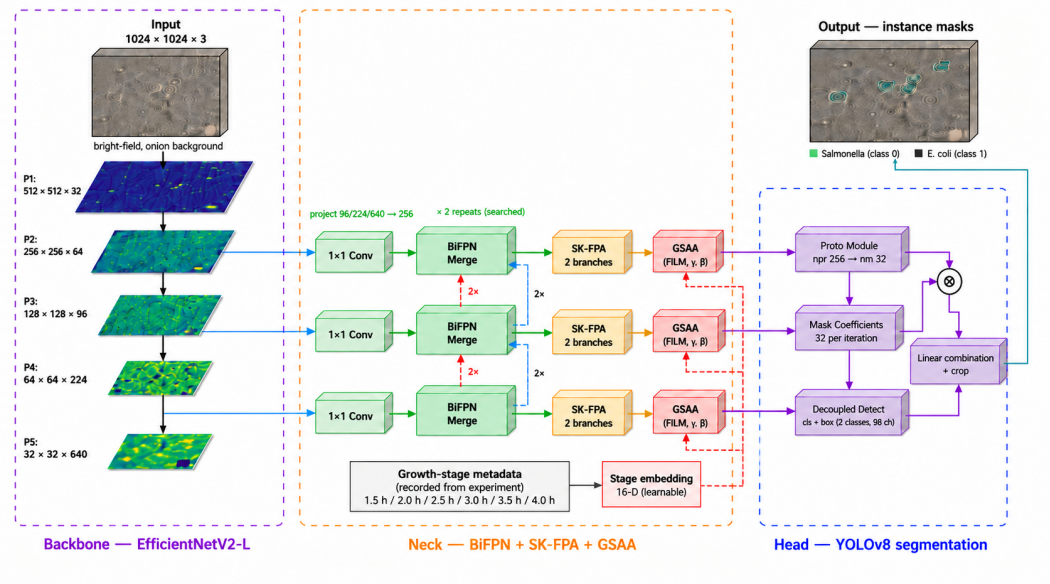


Figure 4. Overview of the proposed BactSegNet architecture. Multi-scale features P3, P4, and P5 from the EfficientNetV2-L backbone (shown as channel-mean activation maps) are projected from their native widths (96, 224, and 640 channels) to a uniform 256 channels and fused by a two-repeat BiFPN neck with top-down and bottom-up pathways. Each level is refined by a Selective-Kernel Feature-Pyramid Attention (SK-FPA) module and modulated by a Growth-Stage-Aware Attention (GSAA) module using a 16-dimensional learnable embedding of the recorded acquisition stage, shared across levels. The YOLOv8 segmentation head combines 32 prototype masks generated from the P3 level with per-location mask coefficients predicted at all three levels to produce the final instance masks for *Salmonella* and *E. coli*. Arrows indicate the direction of feature flow between the depicted components. The configuration shown was selected by BE-NAS.

3.2.3. Weighted Bidirectional FPN (BiFPN-Style) Neck

The neck fuses the three backbone levels so that high-resolution, spatially precise features, and low-resolution, semantically rich features inform one another. We adopt a weighted bidirectional feature-pyramid (BiFPN-style) neck [22]. Because the backbone levels arrive with different channel counts, each level $\mathbf{C}_i$ is first projected to a uniform width C by a 1×1 convolution,

$$\mathbf{P}_i^{(0)} = \mathrm{Conv}_{1\times1}(\mathbf{C}_i), \qquad i \in \{3,4,5\}, \tag{1}$$

so that all levels share the channel dimension C (the search selected $C = 256$). The neck then combines the levels through a top-down pathway that upsamples coarser levels into finer ones and a bottom-up pathway that downsamples finer levels into coarser ones. All fusions use learnable, non-negative, normalised weights (fast normalised fusion). Writing $\mathcal{U}(\cdot)$ and $\mathcal{D}(\cdot)$ for 2× nearest-neighbour upsampling and strided downsampling, and $\phi(\cdot)$ for a depthwise-separable 3×3 convolution with batch normalisation and a SiLU activation, the top-down features are

$$\tilde{\mathbf{P}}_i = \phi\left(\frac{w_1\,\mathbf{P}_i^{(0)} + w_2\,\mathcal{U}(\tilde{\mathbf{P}}_{i+1})}{w_1 + w_2 + \epsilon}\right), \tag{2}$$

and the fused outputs after the bottom-up pass are

$$\mathbf{P}_i^{\text{out}} = \phi\left(\frac{w_1' \tilde{\mathbf{P}}_i + w_2' \mathcal{D}(\mathbf{P}_{i-1}^{\text{out}})}{w_1' + w_2' + \epsilon}\right), \tag{3}$$

where $w_1, w_2, w_1', w_2' \geq 0$ are learnable fusion weights, constrained to be non-negative by a ReLU, $\epsilon = 10^{-4}$ ensures numerical stability, and the boundary levels are taken directly from the corresponding pass. This bidirectional block is repeated N times, where N (set to 2 by the architecture search) is the number of fusion repeats. The BiFPN topology itself is fixed; BE-NAS searches its channel width and repeat count as part of the broader configuration search.

3.2.4. Selective-Kernel Feature-Pyramid Attention (SK-FPA)

The apparent size of bacterial cells varies with species and growth stage, so a fixed convolutional receptive field is a compromise. The SK-FPA module, shown in Figure 5b, adapts the receptive field per channel following the selective-kernel principle [23]. Each fused feature map $\mathbf{P} \in \mathbb{R}^{C \times h \times w}$ is processed by K parallel convolutional branches with different dilation rates, producing $\{\mathbf{U}_k\}_{k=1}^{K}$ (the search selected $K = 2$, with dilation rates 1 and 2). The branches are fused and pooled into a channel descriptor $\mathbf{z} \in \mathbb{R}^C$,

$$\mathbf{U} = \sum_{k=1}^{K} \mathbf{U}_k, \qquad z_c = \frac{1}{h\,w} \sum_{p=1}^{h} \sum_{q=1}^{w} U_{c,p,q}, \tag{4}$$

after which a lightweight transform followed by a per-branch softmax yields attention weights $\{\mathbf{a}_k\}$ with $\sum_k a_{k,c} = 1$ for every channel c. The output recombines the branches under these weights,

$$\mathbf{P}' = \sum_{k=1}^{K} \mathbf{a}_k \odot \mathbf{U}_k, \tag{5}$$

where $\odot$ denotes channel-wise multiplication with spatial broadcasting. The module is applied independently at P3, P4, and P5 and is shape-preserving: $\mathbf{P}'$ has the same dimensions as $\mathbf{P}$. Equations (4) and (5) are stated for general K; in the configuration selected by the search, $K = 2$, and the per-branch softmax reduces exactly to a single gate and its complement, $a_{1,c} = \sigma(u_c)$ and $a_{2,c} = 1 - a_{1,c}$ with u_c the difference of the two branch logits. This is the form drawn in Figure 5b and the form implemented; the two descriptions are the same operation, not two variants of it.

3.2.5. Growth-Stage-Aware Attention (GSAA)

Bacterial morphology changes systematically with incubation time. We therefore use the recorded acquisition time as explicit conditioning metadata. Each image is assigned to one of six stages, $s \in \{1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ h, from its acquisition metadata encoded in the filename. The stage is mapped to an integer index and then to a learnable embedding $\mathbf{e} = E(s) \in \mathbb{R}^d$. The embedding layer is randomly initialised and trained jointly with the segmentation network; it is not a separately pretrained or frozen image encoder. For the selected BactSegNet-L configuration, $d = 16$.

We use feature-wise linear modulation (FiLM) [24]: two lightweight learned projections map the stage embedding to per-channel scale and shift terms,

$$\boldsymbol{\gamma}(\mathbf{e}) = \text{MLP}_\gamma(\mathbf{e}), \qquad \boldsymbol{\beta}(\mathbf{e}) = \text{MLP}_\beta(\mathbf{e}), \tag{6}$$

which modulate each feature map through an affine transform with a residual connection,

$$\mathbf{F}' = \boldsymbol{\gamma}(\mathbf{e}) \odot \mathbf{F} + \boldsymbol{\beta}(\mathbf{e}) + \mathbf{F}. \tag{7}$$

The same stage embedding is shared across P3, P4, and P5, while each level has its own modulation projections; the module is shape-preserving. The residual path permits an identity-like mapping when conditioning is uninformative. During Mosaic augmentation,

source images are sampled from the same growth stage so that the composite retains a valid single stage assignment. At inference, the corresponding acquisition stage must therefore be available as metadata; no stage is inferred from image appearance.

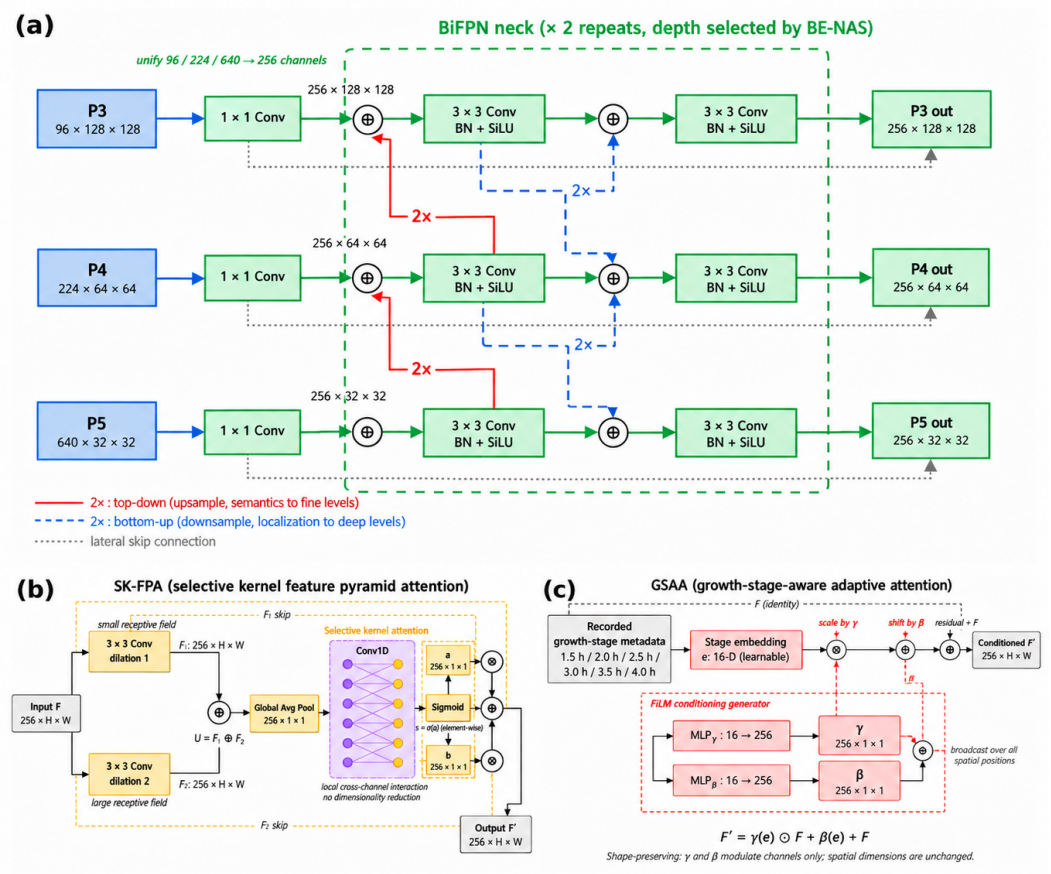


Figure 5. Internal modules of BactSegNet. (**a**) BiFPN projects the three backbone levels to 256 channels and fuses them through repeated top-down and bottom-up pathways; arrows indicate the directions of these pathways. (**b**) SK-FPA combines dilation-1 and dilation-2 branches with channel-wise selective-kernel weighting. (**c**) GSAA maps the recorded acquisition stage to a 16-D learnable embedding and generates FiLM scale and shift terms that modulate each feature-pyramid level through a residual path. The configuration shown is the one selected by BE-NAS.

3.2.6. Segmentation Head

The head follows the YOLOv8 segmentation design [6], which produces instance masks in the YOLACT style. A prototype module takes the highest-resolution level (P3), upsamples it, and generates k prototype masks $\{\mathbf{M}_j\}_{j=1}^{k}$ shared across all instances in the image (the selected configuration uses $k = 32$). In parallel, a decoupled detection branch predicts, at each of the three levels, the object class, a bounding box, and k mask coefficients $\{c_{ij}\}$ at each prediction location; the head is anchor-free, following the YOLOv8 design. The mask for detected instance i is the linear combination of the prototypes weighted by its coefficients, passed through a sigmoid $\sigma(\cdot)$, and cropped to its predicted box b_i,

$$\hat{\mathbf{M}}_i = \mathrm{crop}\Big(\sigma\Big(\textstyle\sum_{j=1}^{k} c_{ij}\,\mathbf{M}_j\Big),\, b_i\Big). \tag{8}$$

We retain the standard YOLOv8 segmentation objective, combining classification, bounding-box regression, distribution focal, and mask losses; the two object classes are *Salmonella* and *E. coli*.

3.2.7. BE-NAS Architecture Search with APED Proxy Evaluation

The configuration of the components above is obtained by a budget-efficient evolutionary architecture search, BE-NAS, summarised in Figure 6. Each candidate is encoded as a chromosome of seven genes: the backbone variant (EfficientNetV2-S, -M, or -L), the number of frozen backbone stages, the BiFPN channel width, the number of BiFPN repeats, the presence of a CBAM block, the number of SK-FPA branches, and the GSAA embedding dimension. With the gene option sets used here, the search space has cardinality

$$|\mathcal{S}| = 3 \times 4 \times 4 \times 4 \times 2 \times 2 \times 3 = 2304 \quad (9)$$

candidate architectures. The scope of the search should be read precisely: the seven genes are capacity and branching settings of established modules—backbone variant, frozen depth, neck width and depth, presence of a CBAM block, number of selective-kernel branches, embedding width—so BE-NAS performs a *configuration* search over a fixed macro-topology, not an operation- or topology-level search of the kind reported for cell-level NAS spaces. We make no claim beyond that. To ensure that fitness differences reflect architecture rather than optimisation noise, all training hyperparameters are held identical across candidates so that only the architecture varies.

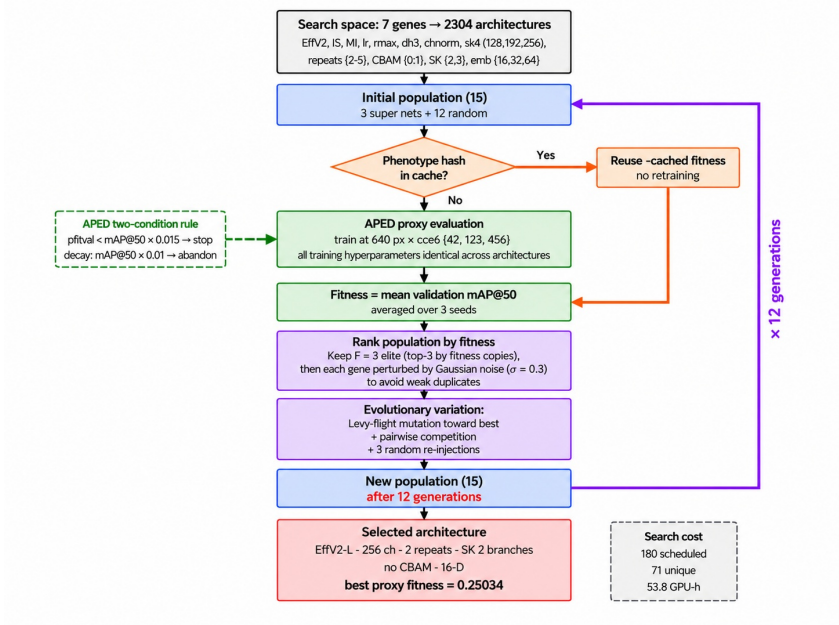


Figure 6. The BE-NAS architecture-search workflow with APED proxy evaluation. A seven-gene chromosome encodes 2304 candidate configurations. Each candidate is scored by a short proxy training run averaged over three seeds, with a phenotype-hash cache reusing stored fitness for recurring architectures; elitism, evolutionary variation (Lévy-flight motion and pairwise competition), and random re-injection drive twelve generations. The search served 180 scheduled candidate evaluations with 71 unique architecture evaluations, corresponding to 213 seed-level proxy runs rather than 540—a 60.6% fewer proxy training runs—at a search cost of 53.8 GPU-hours. This figure covers the architecture search only, not the validation experiments of Section 4.3. Black arrows indicate the main workflow, orange arrows indicate cache reuse, the green dashed arrow indicates the APED stopping rule, and the purple return arrow indicates successive generations.

The search is a generational evolutionary algorithm with a population of $P = 15$ individuals evolved over $G = 12$ generations, initialised with three expert-seeded chromosomes and twelve randomly sampled chromosomes. Each architecture is represented by the seven numerical genes above. Integer genes are decoded by rounding and clipping to their permitted ranges, whereas categorical genes are represented by option indices and decoded by

rounding and clipping to the nearest valid index; consequently, perturbation of a categorical gene, including the backbone gene, always maps to a valid architecture at evaluation time. In each generation, candidates are ranked by fitness, and the $E = 3$ highest-ranked individuals are retained as elites, with each gene perturbed by zero-mean Gaussian noise with standard deviation 0.3 to avoid exact duplication. Additional candidates are generated by a Lévy-flight operator, in which a Lévy step scales the gene-wise difference between two randomly selected population members and is combined with Gaussian jitter, followed by clipping to the valid gene interval. Pairwise competition then retains the fitter of two generated candidates and moves its gene vector a random fraction toward the global-best vector. Three randomly sampled candidates are also re-injected in each new generation to maintain diversity.

Evaluating every candidate by full training would be prohibitively expensive, so BE-NAS scores each candidate with an Adaptive Proxy-Epoch Evaluation (APED) criterion: a short training run at a reduced input resolution of 640 × 640 with batch size 16, retaining the optimiser settings of Table 2, from which the fitness is taken as the mean validation mAP@50 averaged over three random seeds $\mathcal{Z} = \{42, 123, 456\}$ to reduce variance,

$$\mathrm{fit}(a) = \frac{1}{|\mathcal{Z}|} \sum_{z \in \mathcal{Z}} \mathrm{mAP}_{50}^{(z)}(a). \tag{10}$$

The APED rule was defined prospectively with a three-epoch warm-up and a maximum proxy budget of $T_{\max} = 10$ epochs. For a logged validation trajectory, we evaluate the two-epoch change $\Delta_t^{(2)} = \mathrm{mAP}_{50}^{(t)} - \mathrm{mAP}_{50}^{(t-2)}$ and identify a plateau when

$$|\Delta_t^{(2)}| < \varepsilon, \qquad \varepsilon = 0.005, \tag{11}$$

with $n_0 = 3$ and $T_{\max} = 10$. In the reported search, every candidate was trained to the full ten-epoch cap and the rule was applied retrospectively to the complete trajectories, so no candidate-specific early-stopping saving is claimed. The retrospective analysis selected the same early budget for the evaluated candidates, showing that the adaptive rule provided no additional candidate-specific stopping discrimination in this run. We therefore use APED here as a validated low-fidelity proxy-budget criterion rather than as evidence of realised adaptive compute savings. Section 4.3 evaluates whether the resulting early-budget ranking preserves the ordering obtained at the full proxy cap.

The reduction in realised search cost comes from an in-loop cache keyed by a hash $\mathcal{H}(a)$ of the decoded architecture: because elitism and the variation operators cause strong gene combinations to recur, a candidate whose phenotype has already been evaluated reuses its stored fitness without retraining. A total of 25 of the 71 distinct genotypes recurred during the search, and every recurrence returned an identical stored fitness, confirming that the cache is deterministic and introduces no inconsistency between repeated evaluations of the same architecture. Each evaluation comprises one proxy run per seed, so the 180 scheduled evaluations of the full schedule would have required 540 seed-level runs; caching served them with 71 unique architecture evaluations, corresponding to 213 seed-level proxy runs—60.6% fewer proxy training runs than the full schedule would have required—for a total search cost of 53.8 GPU-hours. The configuration selected by the search—an EfficientNetV2-L backbone with no frozen stages, a 256-channel BiFPN neck with two repeats, no CBAM block, two SK-FPA branches, and a 16-dimensional growth-stage embedding—is the BactSegNet model evaluated in the remainder of this paper.

3.3. Implementation and Evaluation Protocol

BactSegNet and the YOLO-family models are implemented in PyTorch 2.5.1 on the Ultralytics framework. All experiments reported in this paper—training, evaluation, and

timing—are conducted on a single compute node equipped with one NVIDIA H100 Tensor Core GPU (NVIDIA Corporation, Santa Clara, CA, USA; Hopper architecture) with 80 GB of HBM3 memory and an NVLink interconnect providing 900 GB/s of bidirectional bandwidth, an Intel Xeon Platinum 8468 (Intel Corporation, Santa Clara, CA, USA; Sapphire Rapids) processor with 48 physical cores and 96 threads (2.10 GHz base, 3.80 GHz turbo), 48 GB of DDR5 system memory, NVMe SSD storage for high-throughput data loading, and Mellanox InfiniBand HDR/NDR node interconnect. The final YOLO-family comparison rerun was executed under Rocky Linux 8/9 with Anaconda3 (2024.02-1), Python 3.11.7, PyTorch 2.5.1+cu121, CUDA 12.1, and Ultralytics v8.4.93, as recorded in the rerun notebook. The recorded BactSegNet implementation/runtime log identifies Ultralytics v8.4.46; we report these run-specific package versions rather than implying a single Ultralytics build across all experiments. The YOLO-family references use the released pretrained checkpoints corresponding to the exact model names reported in Table 4. In particular, YOLO26 and its open-vocabulary extension, YOLOE-26, are distinct released model families [46,47] rather than alternate labels for one checkpoint: the comparison uses the exact released segmentation checkpoint names `yolo26s-seg.pt`, `yolo26m-seg.pt`, `yolo26l-seg.pt` and `yoloe-26s-seg.pt`, `yoloe-26m-seg.pt`, `yoloe-26l-seg.pt`, respectively. Because every model is trained and timed on this same system, all accuracy and speed comparisons reported below are made on identical hardware. Every BactSegNet and YOLO-family model is trained under the single optimisation recipe of Table 2, so performance differences within these comparisons reflect the architecture rather than the training schedule. Each multi-seed result uses three independent random seeds—$\{101, 202, 303\}$ for the YOLOv8 baseline and $\{221, 222, 223\}$ for the BactSegNet variants—and we report the mean over seeds. The thirteen additional YOLO-family models are single runs under the same training and evaluation protocol. Two Mask R-CNN FPN baselines, using ResNet-50 and ResNet-101 backbones, are also evaluated on the same held-out test split as single-run two-stage references. They were initialised from the COCO-pretrained Detectron2 model-zoo checkpoints `mask_rcnn_R_50_FPN_3x` and `mask_rcnn_R_101_FPN_3x`, respectively, and trained at 1024×1024 with a batch size of 4, SGD with a base learning rate of 0.001, approximately 100 epochs, and the standard resize and random-flip augmentation. All test metrics are computed once on the held-out test split of 478 images.

Two further baselines are included to cover paradigms that the YOLO family and Mask R-CNN do not represent; like every other model in this study, both were trained and evaluated on the compute node described above, although their input resolutions and evaluation protocols differ as noted below. The first is RF-DETR-Seg-Large [9], the instance-segmentation variant of a real-time detection transformer with a DINOv2-pretrained backbone, fine-tuned from the released pretrained weights. It was trained at 1008×1008, the nearest valid resolution to the 1024×1024 used elsewhere, since its backbone requires input dimensions divisible by 24; the number of epochs (100), the batch size (8, with gradient accumulation of 2 for an effective batch of 16), the optimiser (AdamW), and the data split were matched to the other models, while the learning rate was set to 1×10^{-4} because the 1×10^{-3} used by the one-stage models diverges for DETR-style transformers. Metrics were computed with the model's native COCO evaluator, and mask (segmentation) AP is reported throughout for comparability. The second is Cellpose-SAM [11], a cellular-segmentation foundation model that adapts the pretrained transformer backbone of the Segment Anything Model [10] into the Cellpose framework. It was fine-tuned—not evaluated zero-shot—from the released `cpsam` weights for approximately 100 epochs on the same split, using its recommended recipe (AdamW, learning rate 1×10^{-5}, weight decay 0.1); the polygon annotations were converted to integer-instance masks, and the conversion was verified by matching per-image mask counts against the label files. Inference used a fixed

cell diameter of 78.3 px, the median ground-truth diameter measured on the validation set, as Cellpose's automatic diameter estimation on the validation set over-segmented and lowered precision. Cellpose-SAM is class-agnostic: it separates cells from background but does not distinguish *Salmonella* from *E. coli*. Cellpose-SAM was evaluated using its native average precision at fixed IoU thresholds, defined as $TP/(TP+FP+FN)$. This metric differs from the confidence-ranked COCO-style average precision used for BactSegNet and the class-aware comparison models. Therefore, Cellpose-SAM is included as a foundation-model reference for segmentation quality rather than as a directly ranked quantitative competitor. Its native AP values are reported separately and are not used to calculate percentage-point differences relative to BactSegNet.

Table 4. Comparison with the baseline and state-of-the-art instance-segmentation models on the held-out test set. The YOLO-family models and BactSegNet were trained under the common optimisation recipe at 1024 × 1024. The YOLOv8s baseline and the three BactSegNet variants are means over three random seeds; the thirteen additional YOLO-family models, the two Mask R-CNN FPN checkpoints, RF-DETR-Seg-L, and Cellpose-SAM are single-run references. All reference models were evaluated on the same held-out test split; their precision and recall were calculated at operating confidence thresholds selected on the validation set. Parameter counts and FLOPs are reported for all models where available. All reported AP values are mask metrics. Seed-level standard deviations in mAP@50 are ±0.0041 (YOLOv8s-seg), ±0.0040 (BactSegNet-S), ±0.0054 (BactSegNet-M), and ±0.0049 (BactSegNet-L). YOLO26 and YOLOE-26 denote distinct released checkpoint families; the row labels reproduce the checkpoint filenames used in the experiments. Best result in each column in bold. See notes below the table for the marked rows.

Model	mAP@50	mAP@50–95	Precision	Recall	Params (M)	FLOPs (G)
YOLOv8s-seg (baseline)	0.9280	0.6711	0.9025	0.9068	11.79	109.3
yolov8m-seg	0.9391	0.6898	0.9185	0.9158	27.24	282.6
yolov8l-seg	0.9402	0.6791	0.9169	0.9194	45.94	565.2
yolov9c-seg	0.9357	0.6851	0.9153	0.9221	27.84	407.2
yolov9e-seg	0.9411	0.6858	0.9191	0.9241	60.45	635.1
yolo11s-seg	0.9373	0.6710	0.9247	0.9196	10.08	91.1
yolo11m-seg	0.9357	0.6765	0.9095	0.9136	22.36	316.4
yolo11l-seg	0.9357	0.6820	0.9250	0.9215	27.62	365.2
yolo26s-seg	0.9400	0.6835	0.9161	0.9066	11.44	101.4
yolo26m-seg	0.9412	0.6985	0.9245	0.9029	26.97	363.4
yolo26l-seg	0.9310	0.6840	0.9001	0.8966	31.38	410.5
yoloe-26s-seg	0.9397	0.6749	0.9242	0.9155	11.44	101.4
yoloe-26m-seg	0.9399	0.6816	0.9156	0.9145	26.97	363.4
yoloe-26l-seg	0.9349	0.6813	0.9287	0.9106	31.38	410.5
Mask R-CNN R50-FPN	0.9163	0.6806	0.9120	0.9170	43.9	167.2
Mask R-CNN R101-FPN	0.9210	0.6850	0.9170	0.9200	62.9	226.7
RF-DETR-Seg-L [a]	0.9368	0.7006	0.9223	0.9336	35.56	163.0
Cellpose-SAM [b]	0.7689	—	0.8636	0.8220	304.6	620.7 [c]
BactSegNet-S	0.9466	0.6989	0.9402	0.9314	51.43	244.1
BactSegNet-M	0.9530	0.7041	**0.9490**	0.9360	83.86	348.5
BactSegNet-L	**0.9636**	**0.7154**	0.9487	**0.9397**	148.12	630.7

[a] RF-DETR-Seg-L was trained and evaluated at 1008×1008, the nearest valid resolution for its backbone, and its efficiency figures were measured at that resolution. [b] Cellpose-SAM performs class-agnostic segmentation and does not distinguish the two species; it reports AP at fixed IoU thresholds rather than COCO mAP@50–95, so its AP@0.50 is listed in the mAP@50 column, the mAP@50–95 entry is left empty, and its row is a reference point rather than a ranked competitor. [c] Cellpose-SAM FLOPs are per 256 × 256 tile, as the model tiles images internally.

We evaluate segmentation quality with the standard COCO-style mask average precision. We report mAP@50 (mask AP at an IoU threshold of 0.50) and mAP@50–95 (averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05), together with precision and recall on the held-out test set using an operating confidence threshold selected on the validation

set as the threshold that maximised F1. The confidence threshold of 0.001 is used only to construct the precision–recall curve for average-precision computation and is not the deployment operating threshold; no test labels were used to select this threshold. Per-class average precision is reported for *Salmonella* and *E. coli* separately. Object sizes are reported in physical units throughout, using the pixel calibration given in Section 3.1.

4. Results

4.1. Comparison with State-of-the-Art Methods

We compare BactSegNet against a strong YOLOv8 segmentation baseline, thirteen recent YOLO-family instance-segmentation models, two standard two-stage mask R-CNN FPN baselines [5], a real-time detection transformer [9], and a cellular-segmentation foundation model [11]. The sixteen class-aware reference models span three architectural paradigms—one-stage CNN detectors, a two-stage CNN detector, and a transformer detector—and Cellpose-SAM is reported separately as a fourth-paradigm, class-agnostic foundation-model reference rather than as a ranked competitor. The YOLOv8s baseline and the BactSegNet variants are multi-seed results. The thirteen additional YOLO-family models are single-run architectural references trained under the common YOLO-family protocol, while the mask R-CNN R50-FPN and R101-FPN checkpoints, RF-DETR-Seg-L, and Cellpose-SAM, are single-run references evaluated on the same held-out test split. Table 4 summarises the comparison. BactSegNet-L attains a mask mAP@50 of 0.9636 and mAP@50–95 of 0.7154. Relative to the multi-seed YOLOv8s baseline (0.9280), this corresponds to a 3.56 pp increase in mAP@50. Comparisons with stronger recent reference architectures show the same ordering: BactSegNet-L is 2.24 pp higher in mAP@50 than yolo26m-seg (0.9412), the strongest single-run comparator on that metric, and exceeds RF-DETR-Seg-L (0.9368/0.7006) by 2.68 pp in mAP@50 and 1.48 pp in mAP@50–95. Among the two-stage references, mask R-CNN R101-FPN is the stronger model, with mAP@50 of 0.9210 and mAP@50–95 of 0.6850; BactSegNet-L exceeds it by 4.26 and 3.04 pp, respectively. At validation-selected operating thresholds, Mask R-CNN R50-FPN records precision and recall of 0.9120 and 0.9170, while R101-FPN records 0.9170 and 0.9200. Its per-class AP@50 values are 0.8990 for *Salmonella* and 0.9430 for *E. coli*, compared with 0.9009 and 0.9317 for mask R-CNN R50-FPN. All three BactSegNet variants report higher mAP@50 values than all sixteen class-aware single-run reference results that report mask mAP@50. The additional single-run models provide broad architectural references, while the YOLOv8s baseline and BactSegNet variants provide the multi-seed comparison. Accordingly, the single-run results are used for model-level comparison rather than statistical significance claims. All class-aware comparison models receive the microscopy image as visual input; BactSegNet additionally uses the recorded acquisition stage as conditioning metadata through GSAA. The ablated configuration without GSAA still reaches an mAP@50 of 0.9445, while the controlled stage-perturbation analysis below isolates the contribution of correct stage assignment without changing model capacity.

To quantify test-set sampling uncertainty, we additionally performed image-level bootstrap resampling with 5,000 iterations. The resulting 95% confidence intervals for mask mAP@50 were 0.9447–0.9717, 0.9239–0.9624, and 0.9342–0.9662 for seeds 221, 222, and 223, respectively.

4.1.1. Comparison with the Transformer Paradigm

RF-DETR-Seg-L is the strongest single-run comparator on mAP@50–95 and reaches mAP@50/mAP@50–95 of 0.9368/0.7006, with precision and recall of 0.9223 and 0.9336. BactSegNet-L reaches 0.9636/0.7154, corresponding to gains of 2.68 pp in mAP@50 and 1.48 pp in mAP@50–95. RF-DETR-Seg-L uses 35.56 M parameters and 163.0 G FLOPs and runs at 90.0 fps (11.11 ms per image), providing a strong efficiency reference alongside

the accuracy comparison. BactSegNet-S reaches 0.9466 mAP@50 and 0.6989 mAP@50–95, further illustrating the accuracy–compute trade-off across model scales; the full BactSegNet results show an accuracy advantage concentrated at the early growth stages and on the target pathogen as Section 4.6 shows. A further parallel is worth noting: RF-DETR is itself obtained by architecture search, so this comparison contrasts two searched designs rather than a searched model against hand-designed ones.

4.1.2. Comparison with a Segmentation Foundation Model

Cellpose-SAM, fine-tuned on our training split, achieves native AP@0.50 = 0.7689, with precision and recall of 0.8636 and 0.8220, respectively, and lower native AP values at stricter overlap thresholds (AP@0.75 = 0.4894 and AP@0.90 = 0.1437). Because Cellpose-SAM's native AP definition differs from the COCO-style AP used for BactSegNet and the class-aware comparison models, these values are presented as a separate segmentation reference and no direct percentage-point performance difference is calculated.

4.1.3. Scaling Behaviour

The three BactSegNet variants follow a clean monotonic ordering L > M > S in both mAP@50 (0.9636 > 0.9530 > 0.9466) and mAP@50–95 (0.7154 > 0.7041 > 0.6989), confirming that the searched configuration scales predictably with backbone capacity. The corresponding efficiency is reported in Table 5: the table now also includes the recorded YOLO-family throughput and latency measurements, including 171.9 fps (5.82 ms/image) for YOLOv8s-seg and 101.7 fps (9.83 ms/image) for yolo26m-seg. The smaller variants trade accuracy for speed in a controlled way, with BactSegNet-S reaching 72.5 fps at 51.43 M parameters while retaining an mAP@50 of 0.9466, which still exceeds every single-run comparison result in mAP@50. The mask R-CNN R50-FPN and R101-FPN baselines run at 36.8 and 28.7 fps, respectively, with latencies of 27.19 and 34.79 ms per image on the same hardware. This gives a practical accuracy–latency range from which a deployment can select according to its throughput requirements. Because every model was timed on the same GPU, these figures are directly comparable, subject only to the differing input resolutions of RF-DETR-Seg-L and the internal tiling of Cellpose-SAM: RF-DETR-Seg-L achieves the highest throughput among the models for which end-to-end speed is reported in Table 5, while Cellpose-SAM is by a wide margin the slowest. It is not the smallest model in this study—yolo11s-seg is, at 10.08 M parameters and 91.1 GFLOPs (Table 4)—so the throughput result should not be read as a parameter-count result.

Table 5. Model size, computational cost, and inference speed of the reference models and BactSegNet variants. All measurements were taken on the same NVIDIA H100 GPU, so the figures are directly comparable. The mask R-CNN baselines and the BactSegNet variants were measured at 1024 × 1024 input; RF-DETR-Seg-L was measured at 1008 × 1008, the nearest valid resolution for its backbone, and Cellpose-SAM at full resolution. Cellpose-SAM FLOPs are per 256 × 256 tile because the model tiles images internally; its speed and latency are end-to-end per image. † BactSegNet-Efficient is the cost-aware configuration introduced in Section 4.9; it is listed here so that all efficiency figures appear in one place.

Model	Params (M)	FLOPs (G)	Speed (fps)	Latency (ms/img)
YOLOv8s-seg	11.79	109.3	171.9	5.82
yolov8m-seg	27.24	282.6	92.0	10.87
yolov8l-seg	45.94	565.2	80.2	12.47
yolov9c-seg	27.84	407.2	79.5	12.58
yolov9e-seg	60.45	635.1	47.0	21.29
yolo11s-seg	10.08	91.1	163.1	6.13
yolo11m-seg	22.36	316.4	96.5	10.37

Table 5. *Cont.*

Model	Params (M)	FLOPs (G)	Speed (fps)	Latency (ms/img)
yolo11l-seg	27.62	365.2	82.2	12.17
yolo26s-seg	11.44	101.4	130.9	7.64
yolo26m-seg	26.97	363.4	101.7	9.83
Mask R-CNN R50-FPN	43.9	167.2	36.8	27.19
Mask R-CNN R101-FPN	62.9	226.7	28.7	34.79
RF-DETR-Seg-L	35.56	163.0	90.0	11.11
Cellpose-SAM	304.6	620.7	1.67	599.57
BactSegNet-S	51.43	244.1	72.5	13.79
BactSegNet-M	83.86	348.5	52.5	19.04
BactSegNet-L	148.12	630.7	37.2	26.90
BactSegNet-Efficient †	142.0	510.0	43.9	22.80

4.2. Independent Mixed-Culture Evaluation

To assess whether frame-level cues alone account for species separation, we applied the trained BactSegNet models without retraining to the independent, expert-annotated mixed-culture set described above. Both species occur within the same field of view and therefore share frame-level acquisition conditions. Across seeds $\{221, 222, 223\}$, mask mAP@50 was 0.8857 ± 0.0206 and mask mAP@50–95 was 0.8139 ± 0.0230. Species-specific mask AP@50 was 0.8994 ± 0.0135 for *Salmonella* and 0.8719 ± 0.0286 for *E. coli*. This additional test indicates that frame-level cues alone do not explain the observed species separation. The evaluation covers only the 3.5 and 4.0 h stages and does not establish performance at earlier mixed-culture stages.

4.3. Validation of the Architecture Search

The search is evaluated on proxy validity, proxy budget, and whether the selected architecture was produced by the evolutionary procedure rather than inherited from its initial seeds.

4.3.1. Proxy–Full Agreement

Twelve architectures spanning the observed proxy-fitness range were retrained under the full protocol and evaluated on the validation split. Proxy and full-training rankings agree strongly: Spearman $\rho = 0.867$ ($p < 0.001$) for mask mAP@50 and $\rho = 0.790$ ($p = 0.0022$) for mask mAP@50–95. Eleven of the twelve fully trained candidates have single-run estimates, so these rank comparisons are descriptive validation of the proxy rather than multi-seed uncertainty estimates. Two of the proxy top-three architectures also appear in the full-training top three. The proxy-ranked winner is also the full-training winner among these twelve candidates.

4.3.2. Proxy Budget and Seed Reliability

Retrospective APED analysis identifies epoch 4 as the earliest common stopping point. Across all 71 unique architectures, the epoch-4 ranking agrees with the epoch-10 ranking at $\rho = 0.87$ (95% CI 0.80–0.92, $p < 0.001$), and, among the twelve fully retrained candidates, it again agrees with full training at $\rho = 0.87$ ($p < 0.001$). Treating the three proxy seeds as repeated measurements gives $\mathrm{ICC}(1,1) = 0.91$ ($F_{70,142} = 31.3$, $p < 0.001$), corresponding to a between-/within-architecture standard-deviation ratio of approximately 3.2. Thus, architecture identity accounts for about 91% of proxy-fitness variance. Epoch 4 would have selected the same architecture at 40% of the ten-epoch proxy budget, but this saving is prospective: the reported search retained all ten epochs.

4.3.3. Convergence and Gene-Level Behaviour

Figure 7a shows that elite fitness rises from 0.219 at initialisation to 0.250 at generation 6 and remains there; the population mean rises from 0.190 to about 0.242, and the proportion with fitness ≥ 0.23 reaches 92–100% across generations 6–11. Periodic random re-injection explains later oscillations in the mean and worst curves. The winner was absent from the initial population and differs from the three expert-seeded architectures on three of seven genes, including removal of CBAM. The seven highest-scoring candidates use EfficientNetV2-L, and no EfficientNetV2-S model enters the top ten. All five multi-valued selected genes lie at an option-set extreme, indicating that the defined search space may be conservatively bounded.

Figure 7b,c summarises the architecture factors. Among adaptively sampled candidates, backbone size and BiFPN width have the strongest descriptive association with fitness; CBAM and GSAA embedding width are weaker. These distributions are not causal estimates because the search is not a balanced factorial experiment. The best backbone–neck combination is EfficientNetV2-L with 256 BiFPN channels; the L/128 value of 0.1534 is an individual training-failure outlier. Overall, BE-NAS evaluates 71 unique architectures, 3.1% of the 2304-configuration space. We use it as the documented design procedure for this architecture and do not claim superiority over alternative search strategies.

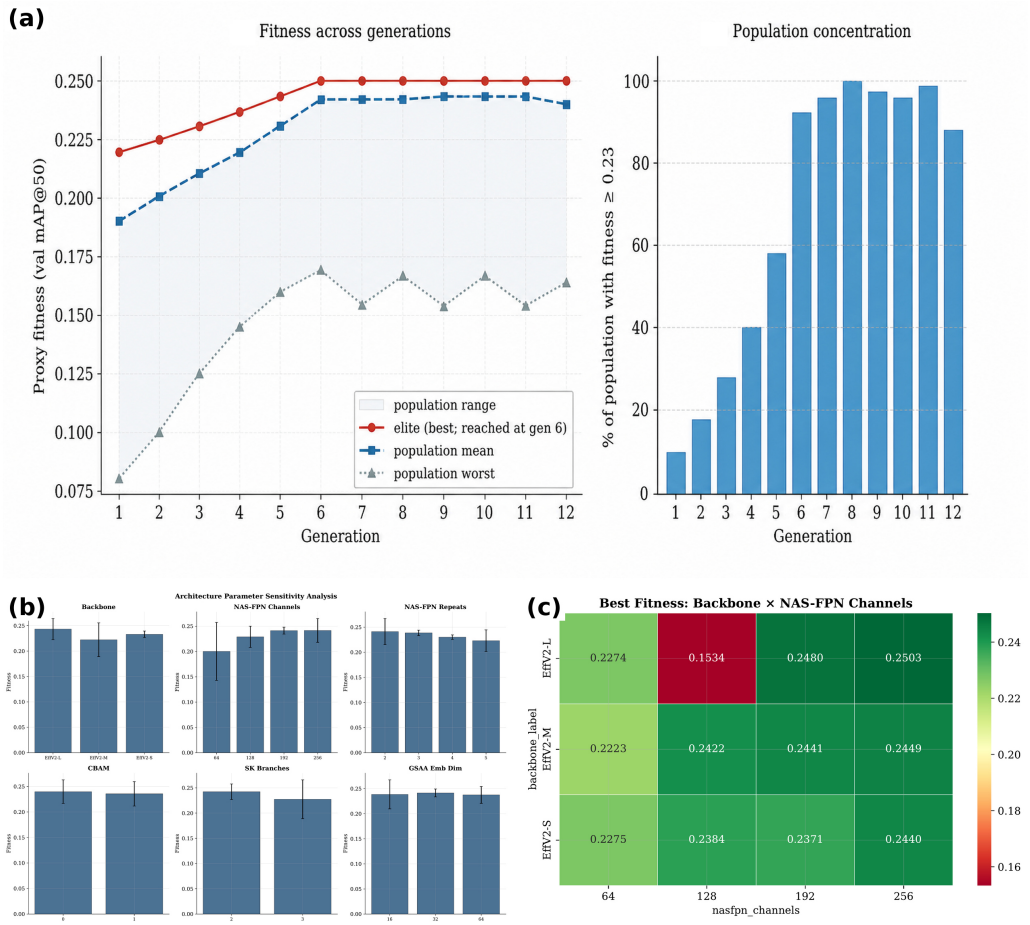


Figure 7. Diagnostics of the BE-NAS search. (**a**) Convergence over twelvegenerations, showing elite, population-level, and strong-candidate behaviour. (**b**) Proxy-fitness distributions across six architecture genes; because sampling is adaptive, these are descriptive rather than causal effects. (**c**) Best proxy fitness for each backbone/BiFPN-width combination, with the selected EfficientNetV2-L/256 configuration highest and the L/128 value (0.1534) an individual training-failure outlier.

4.4. Ablation Study

Each proposed component is removed from BactSegNet-L and retrained with the same three seeds {221, 222, 223}. Table 6 shows consistent drops on both mask metrics. Removing BiFPN has the largest effect (−4.07 pp mAP@50, −2.76 pp mAP@50–95, unadjusted $p = 0.0024$), followed by SK-FPA (−3.01, −1.83 pp, $p = 0.0035$) and GSAA (−1.91, −0.89 pp, $p = 0.0349$). With only three paired seeds, the tests are exploratory; the effect sizes are more informative, exceeding the full-model seed standard deviation (±0.0049 mAP@50) by factors of 8.3, 6.1, and 3.9, respectively.

Table 6. Ablation of the proposed components on BactSegNet-L (mean over three seeds). For w/o BiFPN, multi-scale fusion is replaced by independent 1 × 1 convolution–batch-normalisation–SiLU lateral projections. For w/o SK-FPA, each selective-kernel projection is replaced by a standard 3 × 3 convolution followed by batch normalisation and SiLU. For w/o GSAA, the stage-conditioned modulation is replaced by identity mappings, leaving the feature maps unchanged. The full and ablated configurations use the same seeds {221, 222, 223}. Δ values are changes relative to the full model in percentage points, and p is the unadjusted value from a paired t-test on seed-level mAP@50. Given the three paired seeds, the tests are interpreted as exploratory. Bold entries identify the full-model reference configuration.

Configuration	mAP@50	Δ (pp)	mAP@50–95	Δ (pp)	p-Value
Full BactSegNet-L	**0.9636**	—	**0.7154**	—	—
w/o GSAA	0.9445	−1.91	0.7065	−0.89	0.0349
w/o SK-FPA	0.9335	−3.01	0.6971	−1.83	0.0035
w/o BiFPN	0.9230	−4.07	0.6878	−2.76	0.0024

4.4.1. Stage-Perturbation Control

To distinguish stage-specific conditioning from an effect of additional model capacity, we re-evaluated the same three trained BactSegNet-L checkpoints after randomly shuffling the growth-stage assignments across test images. This control preserves the trained weights, architecture, parameter count, and test images while disrupting the image–stage correspondence. Correct stage assignments yielded 0.9636 ± 0.0049 mAP@50, whereas shuffled assignments yielded 0.9487 ± 0.0040 across seeds {221, 222, 223}, a decrease of 1.49 pp. The result complements the removal ablation by showing that correct stage information contributes to performance even when model capacity is held fixed.

4.4.2. Simple Stage-Conditioning Control

We additionally compared GSAA with a simpler mechanism that receives the same recorded growth-stage information and the same 16-dimensional learned stage embedding but replaces GSAA's stage-dependent scale-and-shift modulation with a single additive linear projection at each of P3–P5. The baseline was retrained under the same data split, training settings, and seeds {221, 222, 223}. Its seed-level mAP@50 values were 0.9570, 0.9590, and 0.9610, giving 0.9590 ± 0.0020 (sample SD). Thus, simple stage conditioning improved mAP@50 by 1.45 pp relative to removing GSAA (0.9445), while the full GSAA model achieved a further 0.46 pp improvement (0.9636). These results indicate that most of the observed gain is associated with providing explicit growth-stage information, while the proposed multi-scale scale-and-shift modulation provides a smaller additional improvement.

4.4.3. Species-Specific Effects

The BiFPN drop is similar for *Salmonella* and *E. coli* (−3.98 and −4.16 pp), whereas SK-FPA (−3.51 vs. −2.51 pp) and especially GSAA (−2.76 vs. −1.07 pp) affect *Salmonella* more strongly (Figure 8). This asymmetry is consistent with the scale statistics in Table 1: median *E. coli* area grows 87.5-fold across the window versus 13.8-fold for *Salmonella*. The

result supports, but does not prove, the intended role of stage conditioning because the components were ablated singly rather than factorially.

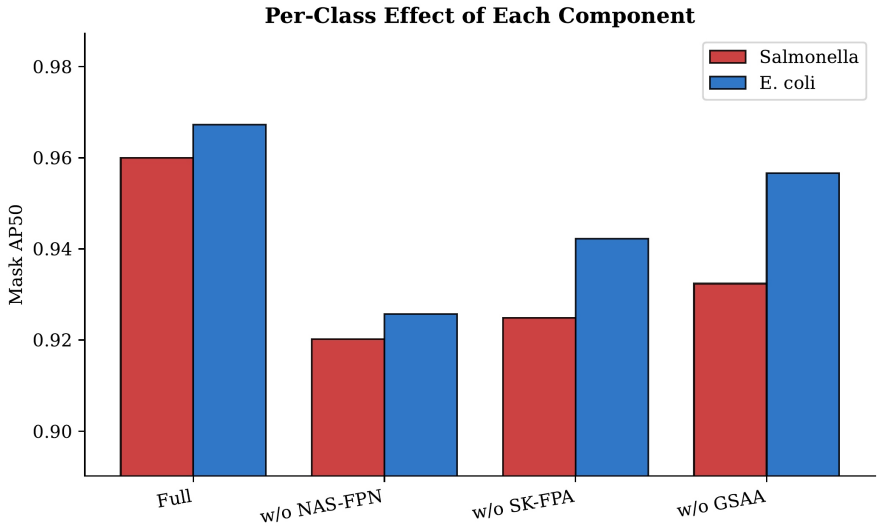


Figure 8. Per-class effect of removing each component from BactSegNet-L (mask AP@50, mean over seeds $\{221, 222, 223\}$). The BiFPN neck contributes almost equally to both organisms, whereas the two task-specific modules contribute asymmetrically: removing SK-FPA and, especially, GSAA costs *Salmonella* substantially more than *E. coli* (-2.76 pp vs. -1.07 pp for GSAA). *Salmonella* is the organism whose object scale changes least over the incubation window and therefore the one whose growth stage is hardest to infer from appearance alone.

4.4.4. Architecture Versus Capacity

Two comparable-capacity controls argue against explaining the gain by size alone. BactSegNet-S (51.43 M parameters, 244.1 GFLOPs) reaches 0.9466 mAP@50 versus 0.9402 for YOLOv8l-seg (45.94 M, 565.2 GFLOPs), a 0.64 pp advantage with 2.3× fewer FLOPs. At comparable compute, BactSegNet-L (630.7 GFLOPs) exceeds YOLOv8l-seg (565.2 GFLOPs) by 2.35 pp, while scaling YOLOv8s to YOLOv8l adds only 1.22 pp. Figure 9 places these controls in the broader comparison: the reference models occupy a shallow accuracy–capacity band, while all three BactSegNet variants lie above it. Because most reference points are single runs from different model families, the plot is descriptive rather than a causal scaling experiment.

4.5. Robustness to the Food-Matrix Background

Because the onion-tissue background is a major observed source of false positives, we evaluate performance separately on onion-present and onion-absent test images. Table 7 reports the result. As expected, both models are more accurate on the cleaner onion-absent images, and BactSegNet improves over the baseline in both regimes: +3.09 pp on onion-absent and +3.65 pp on onion-present images at mAP@50. The margin between the two conditions is small and does not hold uniformly across metrics or against the larger YOLOv8l-seg detector, so we do not claim a general advantage specific to the cluttered background. Resolving the result by species, however, reveals a consistent effect: the improvement on *Salmonella* is markedly larger on onion-present images (+4.90 pp) than on onion-absent images (+3.89 pp), whereas the *E. coli* improvement is essentially flat across the two conditions (+2.31 and +2.38 pp). The benefit of the proposed architecture on the food matrix is therefore concentrated on the target pathogen, which is consistent with the per-stage, per-species analysis in Section 4.6.

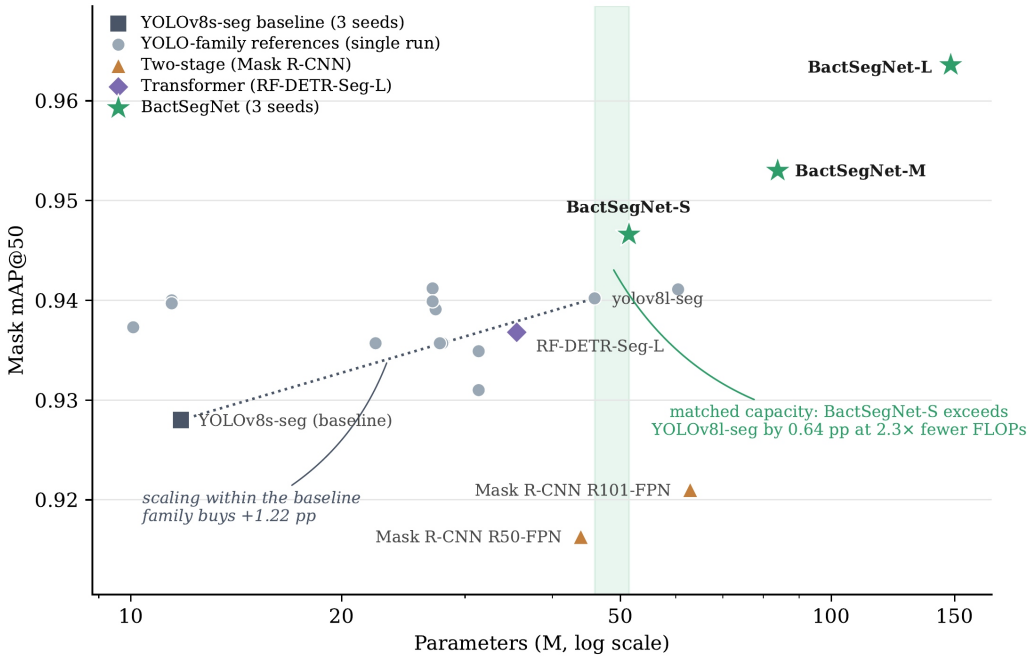


Figure 9. Mask mAP@50 against parameter count for every class-aware model evaluated in this study (values from Table 4; note the logarithmic parameter axis). The reference models form a shallow band in which a 6× range in capacity spans under 2.5 pp of accuracy, and the dotted line shows the +1.22 pp obtained by scaling the baseline family from YOLOv8s-seg to YOLOv8l-seg. All three BactSegNet variants lie above that band: within the shaded comparable-capacity region, BactSegNet-S (51.43 M) exceeds YOLOv8l-seg (45.94 M, a difference of about 12% in parameter count) by 0.64 pp while using 2.3× fewer FLOPs. The vertical offset is consistent with an architectural advantage beyond scaling rather than an isolated measurement of it. Cellpose-SAM is omitted because it performs the easier class-agnostic task and its AP@0.50 = 0.7689 at 304.6 M parameters would compress the vertical axis; its values are given in Table 4. The YOLOv8s-seg baseline and the BactSegNet variants are means over three seeds; all other points are single runs, so the plot is a descriptive comparison rather than a controlled scaling experiment.

Table 7. Segmentation mask mAP@50 by background condition. BactSegNet-L is compared against both the YOLOv8s-seg baseline and the larger YOLOv8l-seg detector on the same test images. *n* is the number of test images in each condition.

Condition	*n*	BactSegNet-L	YOLOv8s	YOLOv8l	Δ vs. YOLOv8s (pp)
Overall	478	0.9636	0.9280	0.9402	+3.56
Onion-absent	237	0.9754	0.9444	0.9505	+3.09
Onion-present	241	0.9480	0.9116	0.9257	+3.65

4.6. Per-Growth-Stage Analysis

Early detection is the most valuable regime for food-safety screening, and it is also the most difficult, since young microcolonies are small and low in contrast. We therefore evaluate accuracy at each of the six growth stages. Figure 10 and Table 8 report mAP@50 per stage for BactSegNet-L and the baseline. These stage-level differences are reported descriptively from the three-seed means and are not treated as statistically supported stage-specific effects. The observed differences are +5.55 pp at 1.5 h, +2.90 pp at 2.0 h, and +6.75 pp at 2.5 h, averaging roughly +5.1 pp over the 1.5–2.5 h window; the observed gap narrows to roughly +0.8 pp over 3.5–4.0 h. The 2.0 h point is a local dip for both models. In addition, the median *Salmonella* microcolony area decreases from 54 μm^2 at 1.5 h to 47 μm^2 at 2.0 h. This non-monotonic change may reflect biological variability across independent experimental plates, differences in microcolony abundance, and background conditions

rather than growth stage alone. Accordingly, the 2.0 h observation is not interpreted as a purely growth-driven morphological transition. The descriptive pattern therefore places the largest observed differences in the early-stage regime, where improved segmentation would be most useful for rapid screening.

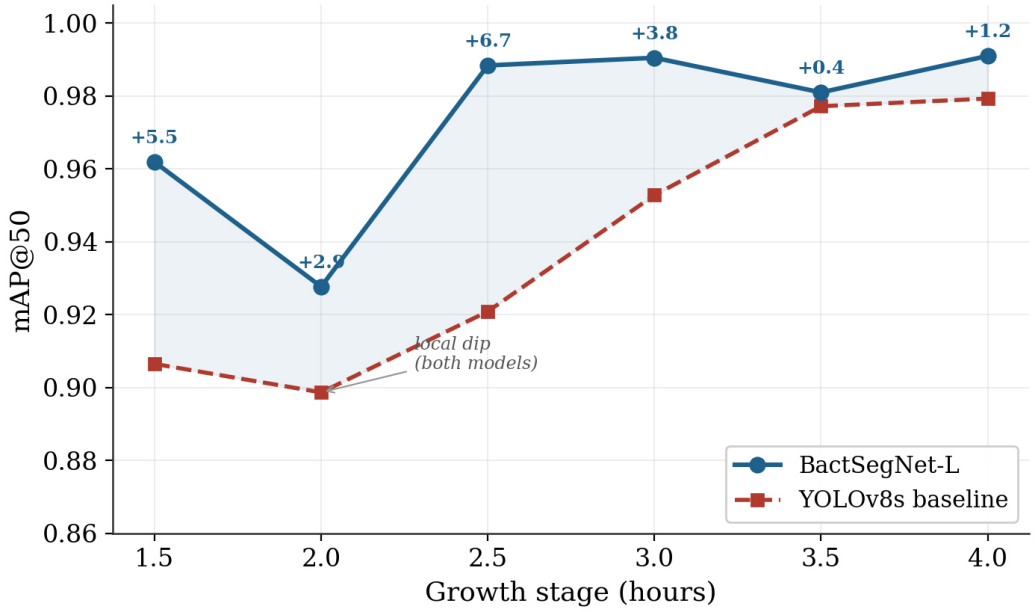


Figure 10. Per-growth-stage mAP@50 for BactSegNet-L and the YOLOv8s baseline (mean over three seeds). The advantage of BactSegNet is largest at the early and intermediate stages (1.5–2.5 h) and narrows at the easier late stages; the 2.0 h point is a local dip for both models. The blue shaded region indicates the performance gap between BactSegNet-L and YOLOv8s across growth stages.

Table 8. Per-growth-stage mAP@50 (BactSegNet-L vs. the YOLOv8s baseline, mean over three seeds). Gains are largest at the early and intermediate stages. Average precision is computed independently within each stage from that stage's own precision–recall curve, so the per-stage values do not average to the overall figures of Table 4. Differences are computed from unrounded values and may differ by up to 0.01 pp from those formed with the rounded figures shown.

Growth Stage	BactSegNet-L	Baseline	Δ (pp)
1.5 h	0.9619	0.9065	+5.55
2.0 h	0.9277	0.8987	+2.90
2.5 h	0.9884	0.9209	+6.75
3.0 h	0.9905	0.9528	+3.77
3.5 h	0.9810	0.9772	+0.38
4.0 h	0.9910	0.9793	+1.18

Per-Species, Per-Stage

Resolving the per-stage result by species shows that the early-stage benefit is strongest and most target-specific for *Salmonella*. At 1.5 h, BactSegNet improves *Salmonella* average precision by +8.66 pp (from 0.8904 to 0.9770), compared with +2.43 pp for *E. coli* at the same stage. The *Salmonella* gains remain positive across the remaining stages (+2.71, +6.69, +3.01, and +1.19 pp at 2.0, 2.5, 3.0, and 4.0 h), with the sole exception of 3.5 h, where the two models are essentially tied (−0.16 pp). The *E. coli* improvements are positive at every stage (+3.08, +6.80, +4.53, +0.92, and +1.16 pp from 2.0 to 4.0 h). These per-species values are descriptive; notably, the largest observed early-stage difference is for *Salmonella* at 1.5 h (+8.66 pp).

4.7. Object-Area Agreement: From Masks to Physical Measurements

Because downstream growth analysis consumes physical size rather than overlap alone, we compute area directly from each predicted mask. Masks produced from the 1024 × 1024 network input are mapped back to the original 2160 × 1620 raster with nearest-neighbour interpolation before pixel counting; ground-truth polygons are rasterised at the same native resolution. This mapping enables physical calibration but does not create native-resolution boundary detail. Pixel count is multiplied by 0.015625 μm^2/px, so the output $\{\text{species}, \text{mask}, \text{area}_{\mu m^2}\}$ requires nolearned area head, additional parameters, or training.

Predictions are matched to ground truth at IoU $\geq$ 0.5. Across 2756 matched test instances from one BactSegNet-L seed, Table 9 reports MAPE, MAE, Pearson correlation, and signed bias; Figure 11 shows the instance-level agreement. Overall MAPE is 12.15%, $r = 0.997$, and bias is +4.35 μm^2; species-specific correlations are 0.999 for *Salmonella* and 0.997 for *E. coli*. The overall correlation spans roughly three orders of magnitude and is therefore dominated by large objects, so it should not be read as near-perfect per-instance segmentation. The analysis is also conditioned on successful matches and is silent about missed instances.

Table 9. Object-area agreement between predicted and ground-truth masks on the test set (2756 matched instances, IoU $\geq$ 0.5, BactSegNet-L at a single seed). MAPE: mean absolute percentage error; MAE: mean absolute error in μm^2; r: Pearson correlation of predicted vs. ground-truth area; bias: mean signed error (positive = over-estimation). Areas are obtained deterministically from the predicted masks by the pixel calibration of Section 3.1; no learned area head is involved. MAPE decreases and MAE increases with growth stage, as expected when relative accuracy improves with object size. The overall Pearson r is computed over instances spanning roughly three orders of magnitude in area and is therefore dominated by the largest objects; the per-stage MAPE is the more informative statistic for the small early-stage instances.

Group	n	MAPE (%)	MAE (μm^2)	r	Bias (μm^2)
Overall	2756	12.15	22.79	0.997	+4.35
1.5 h	814	17.19	9.09	0.948	+3.61
2.0 h	799	16.07	15.85	0.979	−0.15
2.5 h	389	5.92	14.28	0.998	+11.35
3.0 h	302	9.07	19.91	0.999	−4.53
3.5 h	226	3.76	57.68	0.995	−2.27
4.0 h	226	3.35	80.28	0.995	+29.33
Salmonella	1340	12.57	13.25	0.999	+0.69
E. coli	1416	11.75	31.81	0.997	+7.80

Relative error is largest at 1.5 h (17.19%) and falls to 3.35% at 4.0 h, with a local rise to 9.07% at 3.0 h. Early microcolonies occupy a smaller image region, so a boundary deviation of similar absolute size represents a larger fraction of their area. Correspondingly, MAE increases from 9.09 μm^2 at 1.5 h to 80.28 μm^2 at 4.0 h as objects enlarge; against the stage medians of 51 and 1972 μm^2, respectively, the opposite trends of MAPE and MAE are internally consistent.

The same output also supplies per-frame instance count N and total segmented area $\sum_i a_i$. These quantities can support later growth analyses but are not claimed here as calibrated biomass: that would require an independent mapping from projected area to cell mass, such as optical-density or plate-count reference data, which the present acquisition protocol does not provide.

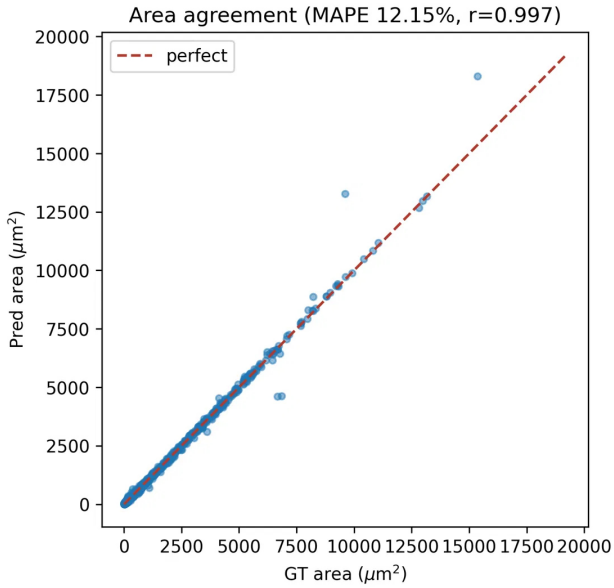


Figure 11. Predicted versus ground-truth object area (μm^2) for the 2756 matched test instances against the identity line (dashed). Each blue dot represents one matched instance. Areas are computed deterministically from the predicted masks using the physical calibration of Section 3.1. The overall Pearson correlation is $r = 0.997$; because the plotted range spans roughly three orders of magnitude, this coefficient reflects fidelity to the overall size distribution and is dominated by the larger instances, and the per-stage MAPE values of Table 9 are the more informative statistic for the small early-stage objects.

4.8. Qualitative Results and Failure Analysis

4.8.1. Successful Predictions

Figure 12 shows representative BactSegNet predictions on test images for both species across the six growth stages. In these examples, every annotated instance is detected and correctly classified, with no false positives, false negatives, or class confusions in the displayed field. The masks follow the irregular object boundaries closely and separate densely clustered microcolonies cleanly, including the small, low-contrast ones at the earliest growth times (1.5–2.0 h), which are least distinguishable from the surrounding medium. The predictions remain stable as the microcolonies enlarge and their morphology becomes more complex at later stages (3.0–4.0 h), confirming that the model generalises across the full temporal range rather than to a single colony size.

4.8.2. Qualitative Comparison with the Foundation-Model Reference

The aggregate gap between BactSegNet and Cellpose-SAM reported in Section 4.1 is concentrated in the regime that matters most for screening. Figures 13 and 14 show Cellpose-SAM's output on three test frames per organism, at growth times 1.5, 2.0, and 2.5 h, drawn from the same images used in Figure 12.

The instance-count deficit is confined to the earliest stage. At 1.5 h, 12 of 17 annotated *E. coli* microcolonies and 4 of 6 annotated *Salmonella* microcolonies are recovered, so roughly 30% of the annotated objects are missed in both fields; the dropped instances are the faint, closely spaced ones inside the dense *E. coli* cluster and the small, low-contrast *Salmonella* rods lying against onion-tissue texture. By 2.0 h, the counts are essentially correct—17 predictions against 16 annotations for *E. coli*, where the surplus instance is a split of an annotated object rather than a newly recovered one, and 9 of 9 for *Salmonella*—and at 2.5 h they match exactly (3 of 3 and 5 of 5). Instance recall alone therefore separates the two models only at the shortest incubation time.

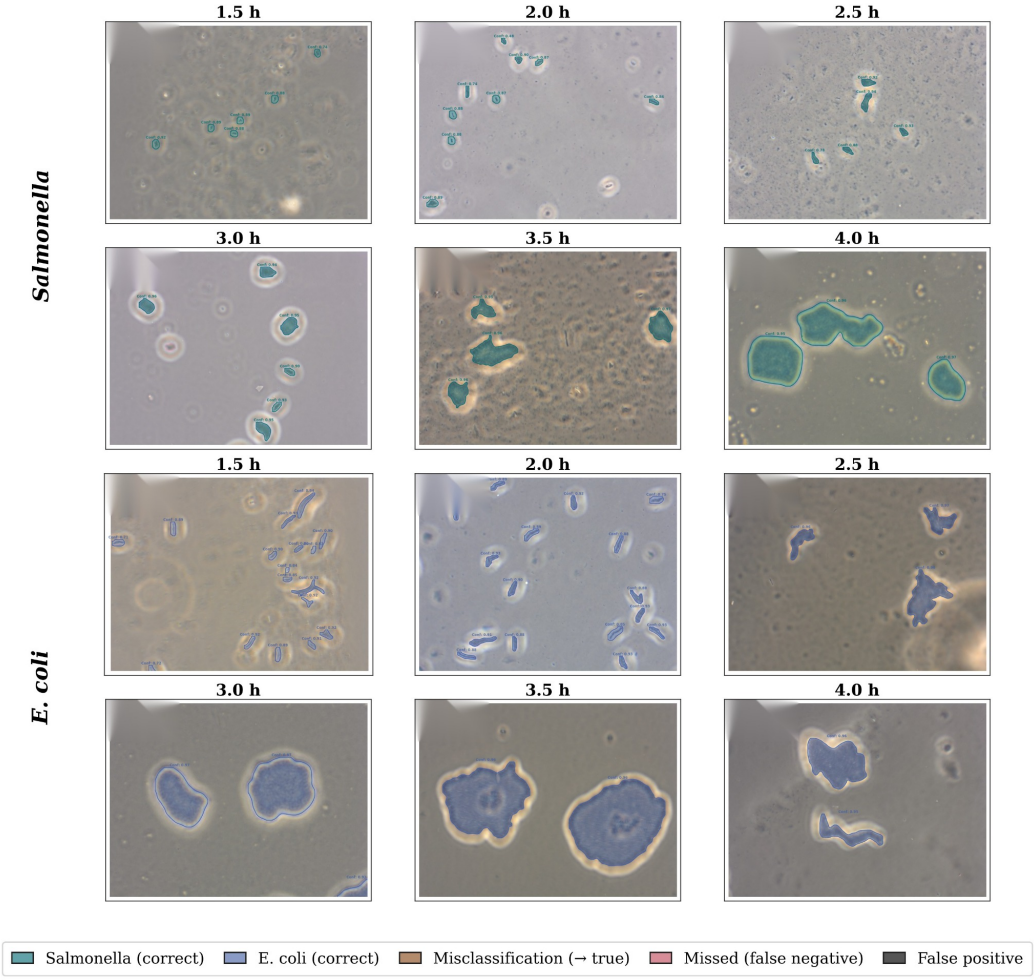


Figure 12. Correct BactSegNet predictions on test images, shown as two 3 × 2 panels ordered by growth time from 1.5 to 4.0 h: *Salmonella* (**top**) and *E. coli* (**bottom**). Teal masks indicate correctly classified *Salmonella*, and blue masks indicate correctly classified *E. coli*; each instance is labelled with its prediction confidence. The legend also lists misclassification, missed instances, and false positives for consistency with the other qualitative panels, but these error categories do not occur in the example fields shown. The panels illustrate the progression from small, low-contrast early-stage microcolonies to larger mature microcolonies; all annotated instances in these fields are detected and classified correctly.

Boundary quality does not follow the same pattern. Wherever Cellpose-SAM retains an instance, the predicted mask is over-dilated relative to the annotated morphology, and the effect persists at 2.0 and 2.5 h where the counts are already correct. It is most pronounced for *Salmonella*, whose thin-rod outline is the cue that separates it from the surrounding onion-tissue texture. This is consistent with the aggregate thresholds in Table 4, where Cellpose-SAM falls from AP@0.50 = 0.7689 to AP@0.75 = 0.4894 and AP@0.90 = 0.1437: its residual error is as much a matter of mask fidelity as of missed detections. BactSegNet detects and correctly classifies every annotated instance in all six of these fields (Figure 12).

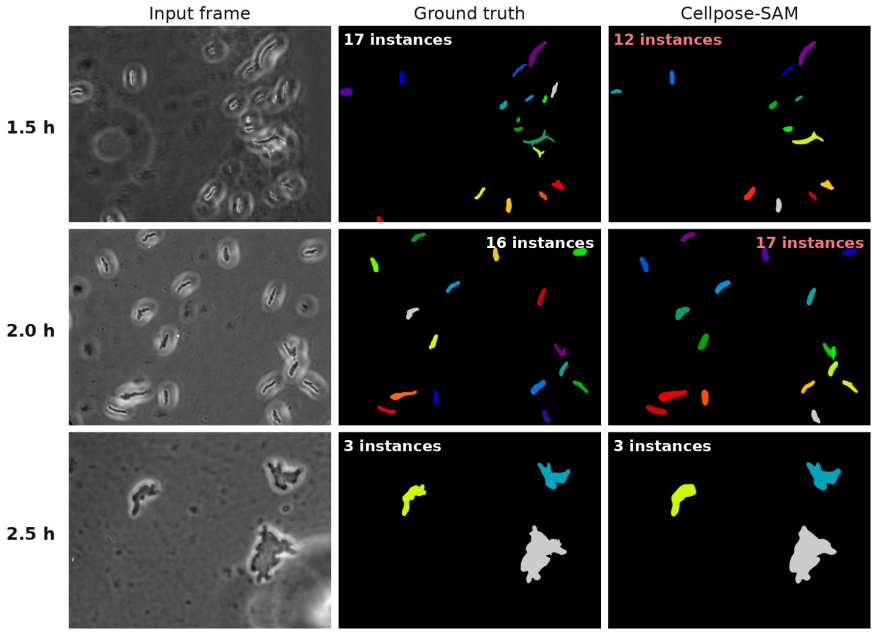


Figure 13. Cellpose-SAM predictions on three *E. coli* test frames, one per growth stage from 1.5 to 2.5 h (rows). Columns show the input frame, the ground-truth instance masks, and the Cellpose-SAM output; mask colour distinguishes instances and does not encode class, as the model is class-agnostic. Rows are labelled by growth time, and the instance count is printed inside each mask panel, in red where the predicted count differs from the annotated one. In total, 12 of 17 annotated microcolonies are recovered at 1.5 h; at 2.0 h, an annotated object is split, giving 17 predictions for 16 annotations; at 2.5 h, all three are recovered. Predicted masks are over-dilated relative to the annotated boundaries at every stage. BactSegNet detects and correctly classifies every annotated instance in all three fields (Figure 12).

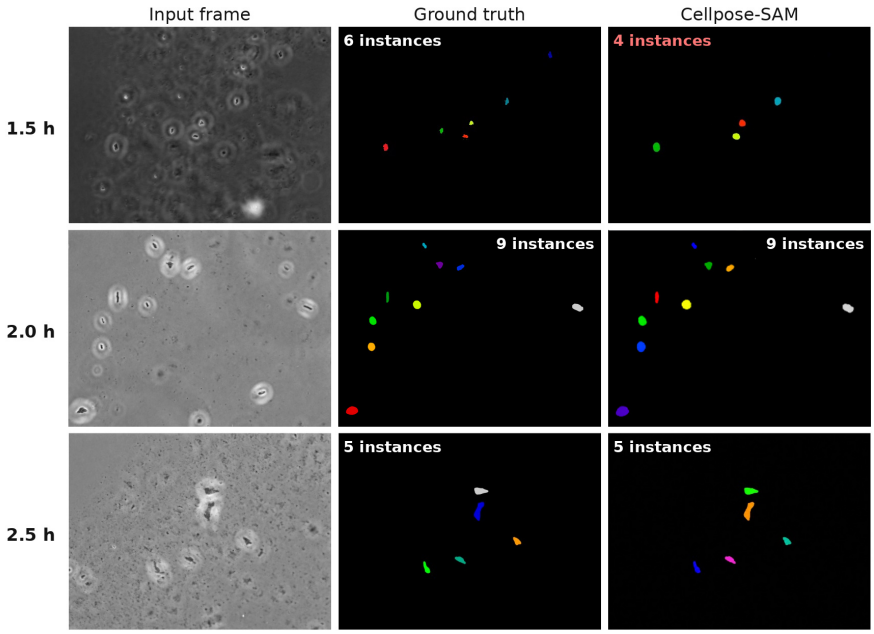


Figure 14. Cellpose-SAM predictions on three *Salmonella* test frames, one per growth stage from 1.5 to 2.5 h (rows), with the same column, row-label, and instance-count conventions as Figure 13. A total of 4 of 6 annotated microcolonies are recovered at 1.5 h, whereas the counts are exact at 2.0 h (9 of 9) and 2.5 h (5 of 5). The retained instances are over-dilated relative to the ground-truth rod morphology, most visibly at 1.5 and 2.0 h, so the thin-rod outline that distinguishes the organism from the surrounding onion-tissue texture is not preserved even where every instance is found. BactSegNet detects and correctly classifies every annotated instance in all three fields (Figure 12).

Taken together, the six frames narrow rather than broaden the qualitative claim. The generalist model becomes competitive on instance count once the microcolonies have grown, and it under-detects specifically at the shortest incubation time—the point at which early screening is most valuable—while its mask boundaries remain coarse at every stage shown. This is qualitative evidence on six frames rather than a quantitative claim; the aggregate figures in Table 4 and the per-stage results of Section 4.6 remain the basis for the comparison.

4.8.3. Failure Analysis

To characterise the model's limitations, we inspected its errors on the test split and grouped them into three types: false negatives (an annotated object is missed), false positives (a background structure is segmented as a microcolony), and misclassifications (a correctly localised object is assigned the wrong species). Figure 15 shows representative examples of each type for both organisms.

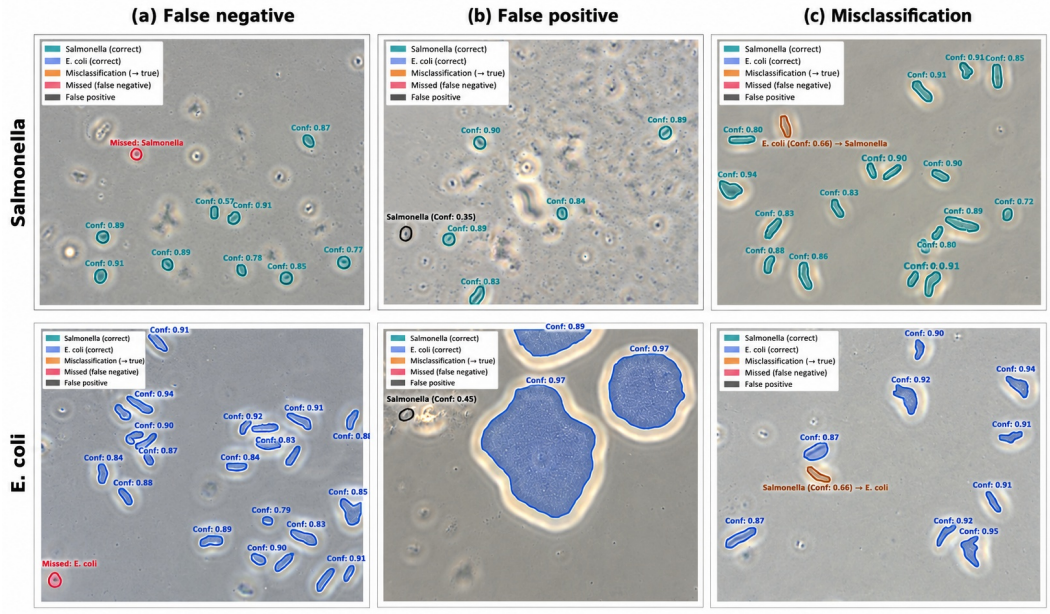


Figure 15. Representative failure modes of BactSegNet on the test split, organised by error type. Columns: false negative (a missed annotated microcolony), false positive (a background structure segmented as a microcolony), and misclassification (a correctly localised instance assigned the wrong species). (**Top row**): *Salmonella* images; (**bottom row**): *E. coli* images. Overlay colour follows the legend (correct *Salmonella*, correct *E. coli*, misclassification, missed/false-negative, false positive). Most errors occur at the earliest growth stage and on onion-tissue background; cross-species false positives are biased toward the *Salmonella* class.

The dominant sources of error are consistent with the difficulty of the imaging setting. False negatives occur almost exclusively at the earliest growth stage (1.5 h), where the target is a small, low-contrast early-stage microcolony that is barely separable from onion tissue or optical artefacts; such instances are the hardest to detect and coincide with the growth stage at which detection is most valuable for early screening. False positives arise where onion-tissue debris, air bubbles, or surface depressions locally resemble a microcolony. Misclassifications are comparatively rare and occur mainly when an object's apparent morphology overlaps between the two species.

A clear asymmetry emerges in the cross-species errors. Across the entire test split, no false positive in a *Salmonella* image was labelled *E. coli*: all spurious detections in *Salmonella* images were themselves labelled *Salmonella*. In contrast, *E. coli* images produced false positives labelled both *E. coli* and, in some cases, *Salmonella* (Figure 15, bottom row). In other words,

the model's spurious detections are biased toward the *Salmonella* class rather than distributed symmetrically between the two organisms. This asymmetry is consistent with the class balance and morphological characteristics of the two organisms in the dataset, and it indicates that the residual false-positive risk is concentrated on, rather than evenly shared across, the two classes. Since every frame was acquired as a pure culture, this asymmetry may partly reflect frame-level appearance cues rather than object morphology, and we therefore draw no operational conclusion from it. Because these errors cluster at the earliest growth stage and on the cluttered food-matrix background, they also identify the two most productive directions for future improvement: higher-resolution or multi-scale handling of small, low-contrast early-stage microcolonies, and stronger background modelling of onion tissue.

4.9. Cost-Aware Architecture Selection

The original search constrained BiFPN repeats to $\{2, 3, 4, 5\}$. The previously reported value of one BiFPN repeat for BactSegNet-Efficient was a typographical error; the correct value is two repeats, which lies within the implemented BE-NAS search space.

$$\mathrm{fit}_\lambda(a) = \mathrm{mAP}_{50}(a) - \lambda \log \mathrm{FLOPs}(a), \tag{12}$$

where the score provides a cost-aware criterion for comparing architectures. The reported BactSegNet-Efficient configuration retains the EfficientNetV2-L backbone while using one frozen backbone stage, a narrower 192-channel BiFPN, two BiFPN repeats, and three SK-FPA branches. The held-out test set was used only for final reporting. Table 10b gives the resulting trade-off. Across three seeds BactSegNet-Efficient reaches 0.9510 ± 0.0008 mAP@50 and 0.7020 ± 0.0008 mAP@50–95, 1.26 and 1.34 pp below BactSegNet-L, at 19.1% fewer FLOPs and 43.9 fps against 37.2 fps. It remains above every class-aware single-run reference in Table 4 on mAP@50.

What is saved is computation rather than model size: parameters fall by only 4.1%, from 148.12 to 142.0 M, since the backbone is unchanged and the neck holds few weights relative to its cost. The neck operates at high spatial resolution—P3 is 128×128 for a 1024×1024 input—so its convolutions are inexpensive in weights and expensive in multiply–accumulates; narrowing it from 256 to 192 channels and halving its depth removes about 4% of the parameters, 19.1% of the FLOPs, and 15.2% of measured latency.

We state the magnitude plainly rather than presenting it as a large efficiency result: a deployment would weigh roughly a fifth less compute against 1.26 pp of mAP@50, which is a narrow rather than a compelling trade. The purpose of the re-ranking is diagnostic. It establishes that the accuracy-only objective, and not the components or the search space, is what placed the widest and deepest neck at the top of the ranking; and the small size of the resulting saving is itself informative, since it indicates that the accuracy–compute trade-off available within these gene ranges is narrow. That is consistent with the earlier observation that all five multi-valued genes were selected at an extreme of their option set, and it strengthens rather than weakens the case for an enlarged search space in future work.

The cost is largest at the early stages and falls monotonically as the colonies enlarge: BactSegNet-Efficient is 1.29 pp below BactSegNet-L at 1.5 h, 1.22 pp at 2.0 h, and between 0.74 and 0.40 pp below it thereafter, while retaining a +4.25 pp advantage over the baseline at 1.5 h (Table 11a). We report the exception symmetrically: at 3.5 h, the lighter configuration falls marginally below the baseline (-0.07 pp), the one stage at which it does not improve on it. Species and background behaviour are retained; for *Salmonella* on onion, the improvement is +3.72 pp versus +4.90 pp for the full model (Table 11b). Area agreement remains similar, at 12.78% MAPE, 23.98 μm^2 MAE, $r = 0.995$, and +5.23 μm^2 bias, compared with 12.15%, 22.79 μm^2, 0.997, and +4.35 μm^2 for BactSegNet-L. Thus, a model at 80.9% of the compute retains 98.7% of the full model's mAP@50. This does not establish a global accuracy–compute frontier because the re-ranking is restricted to phenotypes visited by the accuracy-driven search.

Table 10. Cost-aware architecture and overall trade-off. (**a**) Gene-level comparison of the accuracy-only and cost-aware winners; bold entries differ. (**b**) Held-out test accuracy and cost; accuracy values are means over seeds {221, 222, 223}, and Δ is BactSegNet-Efficient minus BactSegNet-L.

(**a**) Architecture configuration		
Gene	**BactSegNet-L**	**BactSegNet-Efficient**
Backbone variant	EfficientNetV2-L	EfficientNetV2-L
Frozen backbone stages	0	**1**
BiFPN channels	256	**192**
BiFPN repeats	2	2
CBAM block	no	no
SK-FPA branches	2	**3**
GSAA embedding dim.	16	16

(**b**) Accuracy and cost			
Metric	**BactSegNet-L**	**BactSegNet-Efficient**	**Δ**
mAP@50	0.9636 ± 0.0049	0.9510 ± 0.0008	−1.26 pp
mAP@50–95	0.7154 ± 0.0032	0.7020 ± 0.0008	−1.34 pp
Precision	0.9487	0.9385	−1.02 pp
Recall	0.9397	0.9338	−0.59 pp
AP@50 (*Salmonella*)	0.9621	0.9488	−1.33 pp
AP@50 (*E. coli*)	0.9650	0.9532	−1.18 pp
Parameters (M)	148.12	142.0	−4.1%
FLOPs (G)	630.7	510.0	−19.1%
Speed (fps)	37.2	43.9	+18.0%
Latency (ms/img)	26.90	22.80	−15.2%

Table 11. Stratified performance of BactSegNet-Efficient. (**a**) Per-growth-stage mAP@50 compared with BactSegNet-L and the YOLOv8s baseline. (**b**) mAP@50 by species and background against the same baseline. Per-stage AP is computed independently within each stage.

(**a**) By growth stage					
Stage	**BactSegNet-L**	**Efficient**	**Gap**	**Baseline**	**Δ vs. baseline**
1.5 h	0.9619	0.9490	−1.29	0.9065	+4.25
2.0 h	0.9277	0.9155	−1.22	0.8987	+1.68
2.5 h	0.9884	0.9810	−0.74	0.9209	+6.01
3.0 h	0.9905	0.9850	−0.55	0.9528	+3.22
3.5 h	0.9810	0.9765	−0.45	0.9772	−0.07
4.0 h	0.9910	0.9870	−0.40	0.9793	+0.77

(**b**) By species and background			
Condition	**Efficient**	**YOLOv8s**	**Δ vs. baseline**
Overall	0.9510	0.9280	+2.30
Onion-absent	0.9650	0.9444	+2.06
Onion-present	0.9370	0.9116	+2.54
Salmonella (onion-absent)	0.9620	0.9343	+2.77
Salmonella (onion-present)	0.9310	0.8938	+3.72
E. coli (onion-absent)	0.9680	0.9545	+1.35
E. coli (onion-present)	0.9430	0.9294	+1.36

5. Discussion

5.1. What the Results Establish

The experiments support four main conclusions. First, architecture design matters beyond capacity. BactSegNet-S reaches 0.9466 mAP@50 with 51.43 M parameters, exceeding YOLOv8l-seg (0.9402, 45.94 M) while using 2.3× fewer FLOPs; at comparable compute, the architectural gain (2.35 pp) also exceeds the 1.22 pp obtained by scaling YOLOv8s to

YOLOv8l. The ablation identifies BiFPN and SK-FPA as the largest contributors, consistent with the strong scale variation in the data.

Second, low-fidelity NAS should be validated on the target task. Proxy ranking agrees with full training at $\rho = 0.87$, while $\mathrm{ICC}(1,1) = 0.91$ indicates that architecture identity, not seed noise, dominates proxy-fitness variance. Epoch 4 is a retrospectively supported proxy budget for future searches, but the reported search trained every candidate for ten epochs; the realised computational saving came from phenotype caching, which served 180 scheduled evaluations with 71 unique trainings. The cost-aware re-ranking further shows that the accuracy-only objective, rather than the components themselves, drove selection of the largest configuration: at 80.9% of the compute, BactSegNet-Efficient retains 98.7% of the full model's mAP@50. The saving is modest, and we read that as evidence that this gene space offers only a narrow accuracy–compute trade-off rather than as an efficiency claim. Both configurations retain the EfficientNetV2-L backbone, so the trade-off this re-ranking exposes is in the neck rather than in backbone capacity.

Third, the practical gain is concentrated in the difficult operating regime. The improvement over the baseline averages about +5.1 pp over 1.5–2.5 h and about +0.8 pp over 3.5–4.0 h; for *Salmonella*, it reaches +8.66 pp at 1.5 h. The *Salmonella* gain is also larger on onion-present (+4.90 pp) than onion-absent (+3.89 pp) images, although the background effect is not uniform across all metrics or baselines and should not be generalised beyond the reported comparison.

Fourth, calibrated segmentation can produce a physical measurement at no extra learning cost. Across 2756 matched instances, predicted areas achieve 12.15% MAPE and $r = 0.997$, with per-stage MAPE providing the more informative view for small objects. This extends the output from localisation to $\{\text{species}, \text{mask}, \text{area}\}$, which is directly useful for colony-size and growth analyses.

5.2. Comparison Against Other Paradigms

RF-DETR-Seg-L provides the strongest efficiency counterpoint: BactSegNet-L is 2.68 and 1.48 pp higher on mAP@50 and mAP@50–95, but RF-DETR uses 35.56 M parameters and runs at 90.0 fps, so throughput- or memory-constrained deployment may favour the transformer. Cellpose-SAM, fine-tuned on the same split, reaches native AP@0.50 = 0.7689 and falls to AP@0.75 = 0.4894 and AP@0.90 = 0.1437. Because its native AP definition differs from the COCO-style AP used for BactSegNet, and because it is class-agnostic, these values are treated as a separate reference point rather than as a direct ranked comparison. Its qualitative failures at 1.5 h nevertheless show that large-scale generalist pretraining does not automatically solve dense, low-contrast bacterial segmentation in this domain.

5.3. Interpreting the Growth-Stage Conditioning

GSAA explicitly uses the recorded acquisition stage as conditioning metadata. The stage index is mapped to a learned embedding and injected into the segmentation hierarchy through FiLM. Removing it still leaves 0.9445 mAP@50, above every class-aware single-run reference. Its removal affects *Salmonella* 2.6× more than *E. coli*, consistent with useful growth-related conditioning, and the residual path permits an identity-like mapping when the conditioning is uninformative. The controlled stage-perturbation experiment further shows that disrupting image–stage correspondence while preserving architecture, weights, and parameter count reduces mAP@50 by 1.49 pp. In the additional simple-conditioning control, using the same stage embedding with additive projection achieved 0.9590 ± 0.0020 mAP@50, compared with 0.9636 ± 0.0049 for GSAA and 0.9445 without GSAA. This indicates that explicit stage information accounts for most of the conditioning gain, while GSAA's multi-scale scale-and-shift modulation provides a smaller additional

improvement. We therefore interpret GSAA as a stage-conditioned modulation mechanism rather than implying that conditioning is universally beneficial or that all gains arise from the GSAA formulation itself.

5.4. Limitations and Scope

5.4.1. Acquisition Scope

The data come from a single microscope, a single magnification, one food matrix, and two organisms. Generalisation to other imaging setups, media, matrices, species, or growth-stage grids is not established. We searched for a publicly available external dataset with sufficiently comparable bacterial microcolony characteristics and acquisition conditions but did not identify a suitable dataset for direct external validation. One closely related dataset was identified; after contacting its authors, we were informed that it remains part of ongoing research and is not currently available for external use.

5.4.2. Pure-Culture Frames and Species Discrimination

The original training, validation, and test partitions contain one organism per frame. The acquisition design spans 20 onion samples, three replicate plates per sample, and three independent experimental days, with growth stages obtained from separate experimental plates. Although image identifiers do not overlap across partitions, images from the same biological sample, plate, or acquisition session may occur in different partitions; grouping-level independence cannot be retrospectively verified. The independent mixed-culture evaluation (Section 4.2) provides an additional test of species discrimination with both organisms in the same field of view but covers only 3.5 and 4.0 h. The false-positive asymmetry in the original pure-culture test remains a relevant limitation.

5.4.3. Statistical Power

Multi-seed experiments use three seeds, so paired ablation tests are exploratory. Proxy–full agreement is estimated from twelve fully retrained architectures and consequently has a wide interval (95% CI 0.57–0.97 for ρ). The three model seeds represent independent training runs and quantify computational variability. Test-set sampling uncertainty is additionally quantified by the 5000-iteration image-level bootstrap analysis reported above. The biological acquisition itself included 20 onion samples, three replicate plates per sample, and experiments conducted across three independent days.

5.4.4. Search Scope

BE-NAS searches 2304 configurations over seven genes within a fixed macro-topology and is not compared with alternative search strategies. Cost-aware re-ranking is restricted to the 71 phenotypes visited by the accuracy-driven search, so it does not identify a global Pareto frontier.

5.4.5. Scope of the Area Analysis

Area agreement uses successfully matched instances at IoU $\geq$ 0.5 and one seed; it therefore measures mask fidelity, not overall detection accuracy. Mapping predictions back to 2160 × 1620 supports physical calibration but does not restore boundary detail lost at the 1024 × 1024 network representation.

5.4.6. Data and Code Availability

The source code, trained model, configurations, and expert-verified annotated dataset are currently part of ongoing studies and are therefore not publicly available at this stage.

These resources are planned to be made publicly available upon completion of the ongoing studies. Back Matter states the current availability position.

5.5. Future Work

Priority extensions are higher-resolution or explicitly multi-scale treatment of 1.5 h microcolonies, stronger modelling of onion-tissue distractors, factorial interaction studies of the proposed modules, a prospective search using the measured $T_{\max} = 4$ proxy budget, and co-contaminated images with instance-level labels. Independent datasets, additional pathogens, and other food matrices are needed to test whether metadata-conditioned growth-stage modelling generalises beyond the present protocol.

6. Conclusions

We introduced BactSegNet, a one-stage instance-segmentation model for *Salmonella* and *E. coli* microcolonies under the pure-culture bright-field protocol studied here. The model combines EfficientNetV2, BiFPN, SK-FPA, and metadata-conditioned growth-stage-aware FiLM conditioning, with its configuration selected by BE-NAS from a 2304-configuration search space after evaluating 71 unique architectures. Proxy ranking agrees with full training at $\rho = 0.87$ (95% CI 0.57–0.97), architecture identity accounts for 91% of proxy-fitness variance ($\mathrm{ICC}(1,1) = 0.91$), and phenotype caching reduces 180 scheduled evaluations to 71 unique trainings at a total search cost of 53.8 GPU-hours. Retrospective APED analysis identified a common early proxy budget across candidates; because it did not provide candidate-specific stopping discrimination in this run, we do not claim realised adaptive compute savings. The reported search used all ten proxy epochs.

BactSegNet-L achieves 0.9636 mask mAP@50 and 0.7154 mAP@50–95, 3.56 pp above the multi-seed image-only YOLOv8 baseline on mAP@50. Of this gain, 1.65 pp is achieved without GSAA and a further 1.91 pp with GSAA; the simple stage-conditioning result (0.9590 versus 0.9636) indicates that explicit stage information explains most of the conditioning gain. Recorded stage metadata must be available at inference. Its largest gains occur where early screening is most difficult, reaching +8.66 pp for *Salmonella* at 1.5 h and +4.90 pp for *Salmonella* on onion. Across 2756 matched instances, calibrated mask area achieves 12.15% MAPE and $r = 0.997$, providing a deterministic $\{\text{species}, \text{mask}, \text{area}_{\mu m^2}\}$ output without an additional learned head. BactSegNet-Efficient reaches 0.9510 mAP@50 with 19.1% fewer FLOPs and 43.9 fps, providing a more efficient operating point.

The main bounds are equally important: the primary dataset uses one imaging setup, two organisms, pure-culture frames, three-seed experiments, and a defined search space. An additional independent, expert-annotated mixed-culture test at 3.5 and 4.0 h yielded 0.8857 ± 0.0206 mask mAP@50 across three seeds without retraining; earlier mixed-culture stages were not evaluated. Within those limits, the results support searched, scale-adaptive segmentation that uses explicit recorded growth-stage context and reports stratified, physically interpretable outputs rather than aggregate overlap metrics alone.

Author Contributions: Conceptualisation, M.R. and A.A. (Anas AlSobeh); methodology, M.R.; software, M.R.; validation, M.R., A.A. (Anas AlSobeh), N.D. and A.A. (Amer AbuGhazaleh); formal analysis, M.R.; investigation, M.R.; resources, A.A. (Amer AbuGhazaleh); data curation, M.R. and N.D.; writing—original draft preparation, M.R.; writing—review and editing, K.R.A.; visualisation, M.R.; supervision, K.R.A., A.A. (Anas AlSobeh) and A.A. (Amer AbuGhazaleh); project administration, A.A. (Anas AlSobeh) and A.A. (Amer AbuGhazaleh); funding acquisition, A.A. (Anas AlSobeh) and A.A. (Amer AbuGhazaleh). All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the United States Department of Agriculture, National Institute of Food and Agriculture (USDA-NIFA), under the Capacity Building Grants for Non-Land-Grant Colleges of Agriculture program, grant number 2024-02854.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The source code, trained model, configurations, expert-verified instance-segmentation annotations, image-level split definitions, and associated dataset used in this study are currently part of ongoing studies and are therefore not publicly available at this stage. These resources are planned to be made publicly available upon completion of the ongoing studies. The raw bright-field imagery forms part of an evolving microscopy corpus that has also supported previously published studies with different task formulations and annotation regimes [16–18]. The present study uses the re-curated polygon annotations, coverage, and split definitions described in Section 3.1.

Acknowledgments: The authors used Claude (Anthropic, Claude Opus 5.5) to improve the language, grammar, and readability of the manuscript. The authors reviewed and approved the final content and take full responsibility for its accuracy.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Havelaar, A.H.; Kirk, M.D.; Torgerson, P.R.; Gibb, H.J.; Hald, T.; Lake, R.J.; Praet, N.; Bellinger, D.C.; de Silva, N.R.; Gargouri, N.; et al. World Health Organization Global Estimates and Regional Comparisons of the Burden of Foodborne Disease in 2010. *PLoS Med*. **2015**, *12*, e1001923. [CrossRef] [PubMed]
2. Centers for Disease Control and Prevention. Multistate Foodborne Enteric Disease Outbreaks, 2024. Available online: https://www.cdc.gov/foodnet/reports/preliminary-data.html (accessed on 25 August 2026).
3. Center for Infectious Disease Research and Policy (CIDRAP). Illnesses from Contaminated Food Increased in 2024, Severe Cases Doubled. 2025. Available online: https://www.cidrap.umn.edu/foodborne-disease/ (accessed on 25 August 2026).
4. Maturin, L.; Peeler, J.T. BAM Chapter 3: Aerobic Plate Count. In *Bacteriological Analytical Manual*, 8th ed.; Revision, A., Ed.; U.S. Food and Drug Administration: Silver Spring, MD, USA, 2001.
5. He, K.; Gkioxari, G.; Dollár, P.; Girshick, R. Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV), Venice, Italy, 22–29 October 2017*; IEEE: New York, NY, USA, 2017; pp. 2961–2969. [CrossRef]
6. Jocher, G.; Chaurasia, A.; Qiu, J. Ultralytics YOLOv8. 2023. Available online: https://github.com/ultralytics/ultralytics (accessed on 25 August 2026).
7. Wang, C.Y.; Yeh, I.H.; Liao, H.Y.M. YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information. In *Proceedings of the Computer Vision—ECCV 2024, Milan, Italy, 29 September–4 October 2024*; Lecture Notes in Computer Science; Springer: Berlin/Heidelberg, Germany, 2024. [CrossRef]
8. Carion, N.; Massa, F.; Synnaeve, G.; Usunier, N.; Kirillov, A.; Zagoruyko, S. End-to-End Object Detection with Transformers. In *Proceedings of the European Conference on Computer Vision (ECCV), Glasgow, UK, 23–28 August 2020*; Springer: Berlin/Heidelberg, Germany, 2020; pp. 213–229.
9. Robinson, I.; Robicheaux, P.; Popov, M.; Ramanan, D.; Peri, N. RF-DETR: Neural Architecture Search for Real-Time Detection Transformers. In Proceedings of the International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, 23–27 April 2026. [CrossRef]
10. Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A.C.; Lo, W.Y.; et al. Segment Anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 1–6 October 2023*; IEEE: New York, NY, USA, 2023; pp. 4015–4026.
11. Pachitariu, M.; Rariden, M.; Stringer, C. Cellpose-SAM: Superhuman Generalization for Cellular Segmentation. *bioRxiv* **2025**. [CrossRef]
12. Kang, M.; Ting, C.M.; Ting, F.F.; Phan, R.C.W. ASF-YOLO: A Novel YOLO Model with Attentional Scale Sequence Fusion for Cell Instance Segmentation. *Image Vis. Comput*. **2024**, *147*, 105057. [CrossRef]
13. Schmidt, U.; Weigert, M.; Broaddus, C.; Myers, G. Cell Detection with Star-Convex Polygons. In *Proceedings of the Medical Image Computing and Computer Assisted Intervention—MICCAI 2018, 21st International Conference, Granada, Spain, 16–20 September 2018*; Springer: Berlin/Heidelberg, Germany, 2018; Proceedings, Part II; Volume 11071; Lecture Notes in Computer Science; pp. 265–273. [CrossRef]
14. Cutler, K.J.; Stringer, C.; Lo, T.W.; Rappez, L.; Stroustrup, N.; Peterson, S.B.; Wiggins, P.A.; Mougous, J.D. Omnipose: A High-Precision Morphology-Independent Solution for Bacterial Cell Segmentation. *Nat. Methods* **2022**, *19*, 1438–1448. [CrossRef] [PubMed]

15. Panigrahi, S.; Murat, D.; Le Gall, A.; Martineau, E.; Goldlust, K.; Fiche, J.B.; Rombouts, S.; Nöllmann, M.; Espinosa, L.; Mignot, T. MiSiC, a General Deep Learning-Based Method for the High-Throughput Cell Segmentation of Complex Bacterial Communities. *eLife* **2021**, *10*, e65151. [CrossRef] [PubMed]
16. Koirala, B.; AlSobeh, A.M.; Dhahir, N.; AbuGhazaleh, A. Background-Aware Instance Segmentation for Early Detection of *E. coli* and *Salmonella* in Time-Stamped Microscopy Images. In *Proceedings of the 2025 International Conference on Machine Learning and Applications (ICMLA), Boca Raton, FL, USA, 3–5 December 2025*; IEEE: New York, NY, USA, 2025; pp. 1354–1361. [CrossRef]
17. Rababa, M.; AlSobeh, A.; Dhahir, N.; AbuGhazaleh, A. Backbone and Feature Fusion Design for YOLOv8-Based Bacterial Microcolony Detection in Microscopy Images. *Appl. Sci.* **2026**, *16*, 4241. [CrossRef]
18. AlSobeh, A.; AbuGhazaleh, A.; Dhahir, N.; Rababa, M. XAIPath: Temporal-Environmental Explainable AI Framework for Co-Contaminated Food Pathogen Detection in Microscopic Imaging. In *Proceedings of the 54th International Conference on Parallel Processing Companion (ICPP Companion), San Diego, CA, USA, 8–11 September 2025*; ACM: New York, NY, USA, 2025. [CrossRef]
19. White, C.; Safari, M.; Sukthanker, R.; Ru, B.; Elsken, T.; Zela, A.; Dey, D.; Hutter, F. Neural Architecture Search: Insights from 1000 Papers. *arXiv* **2023**, arXiv:2301.08727. [CrossRef]
20. Abdelfattah, M.S.; Mehrotra, A.; Dudziak, Ł.; Lane, N.D. Zero-Cost Proxies for Lightweight NAS. In Proceedings of the 9th International Conference on Learning Representations (ICLR), Virtual, 3–7 May 2021.
21. Tan, M.; Le, Q.V. EfficientNetV2: Smaller Models and Faster Training. In *Proceedings of the 38th International Conference on Machine Learning (ICML), Virtual, 18–24 July 2021*; PMLR (Proceedings of Machine Learning Research): Cambridge, MA, USA, 2021; pp. 10096–10106.
22. Tan, M.; Pang, R.; Le, Q.V. EfficientDet: Scalable and Efficient Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Seattle, WA, USA, 13–19 June 2020*; IEEE: New York, NY, USA, 2020; pp. 10781–10790.
23. Li, X.; Wang, W.; Hu, X.; Yang, J. Selective Kernel Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA, 15–20 June 2019*; IEEE: New York, NY, USA, 2019; pp. 510–519. [CrossRef]
24. Perez, E.; Strub, F.; de Vries, H.; Dumoulin, V.; Courville, A. FiLM: Visual Reasoning with a General Conditioning Layer. In *Proceedings of the AAAI Conference on Artificial Intelligence, New Orleans, LA, USA, 2–7 February 2018*; AAAI Press: Washington, DC, USA, 2018; Volume 32. [CrossRef]
25. Beznik, T.; Smyth, P.; de Lannoy, G.; Lee, J.A. Deep Learning to Detect Bacterial Colonies for the Production of Vaccines. *Neurocomputing* **2022**, *470*, 427–431. [CrossRef]
26. Makrai, L.; Fodróczy, B.; Nagy, S.Á.; Czeiszing, P.; Csabai, I.; Szita, G.; Solymosi, N. Annotated Dataset for Deep-Learning-Based Bacterial Colony Detection. *Sci. Data* **2023**, *10*, 497. [CrossRef] [PubMed]
27. Whipp, J.; Dong, A. YOLO-based Deep Learning to Automated Bacterial Colony Counting. In *Proceedings of the 2022 IEEE Eighth International Conference on Multimedia Big Data (BigMM), Naples, Italy, 5–7 December 2022*; IEEE: New York, NY, USA, 2022; pp. 120–124. [CrossRef]
28. Kang, R.; Park, B.; Eady, M.; Ouyang, Q.; Chen, K. Single-Cell Classification of Foodborne Pathogens Using Hyperspectral Microscope Imaging Coupled with Deep Learning Frameworks. *Sens. Actuators B Chem.* **2020**, *309*, 127789. [CrossRef]
29. Bhattacharya, S.; Wasit, A.; Earles, J.M.; Nitin, N.; Yi, J. Enhancing AI microscopy for foodborne bacterial classification using adversarial domain adaptation to address optical and biological variability. *Front. Artif. Intell.* **2025**, *8*, 1632344. [CrossRef] [PubMed]
30. Nagy, S.Á.; Makrai, L.; Csabai, I.; Tőzsér, D.; Szita, G.; Solymosi, N. Bacterial Colony Size Growth Estimation by Deep Learning. *BMC Microbiol.* **2023**, *23*, 307. [CrossRef] [PubMed]
31. Stringer, C.; Wang, T.; Michaelos, M.; Pachitariu, M. Cellpose: A Generalist Algorithm for Cellular Segmentation. *Nat. Methods* **2021**, *18*, 100–106. [CrossRef] [PubMed]
32. Lin, T.Y.; Dollár, P.; Girshick, R.; He, K.; Hariharan, B.; Belongie, S. Feature Pyramid Networks for Object Detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017*; IEEE: New York, NY, USA, 2017; pp. 2117–2125. [CrossRef]
33. Liu, S.; Qi, L.; Qin, H.; Shi, J.; Jia, J. Path Aggregation Network for Instance Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018*; IEEE: New York, NY, USA, 2018; pp. 8759–8768. [CrossRef]
34. Ghiasi, G.; Lin, T.Y.; Le, Q.V. NAS-FPN: Learning Scalable Feature Pyramid Architecture for Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA, 15–20 June 2019*; IEEE: New York, NY, USA, 2019; pp. 7036–7045. [CrossRef]
35. Hu, J.; Shen, L.; Sun, G. Squeeze-and-Excitation Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018*; IEEE: New York, NY, USA, 2018; pp. 7132–7141. [CrossRef]
36. Woo, S.; Park, J.; Lee, J.Y.; Kweon, I.S. CBAM: Convolutional Block Attention Module. In *Proceedings of the European Conference on Computer Vision (ECCV), Munich, Germany, 8–14 September 2018*; Springer: Berlin/Heidelberg, Germany, 2018; pp. 3–19. [CrossRef]

37. Kong, Y.; Shen, Z. Microorganism Detection in Activated Sludge Microscopic Images Using Improved YOLO. *Appl. Sci.* **2023**, *13*, 12406. [CrossRef]
38. Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; et al. DINOv2: Learning Robust Visual Features without Supervision. *arXiv* **2024**, arXiv:2304.07193.
39. Elsken, T.; Metzen, J.H.; Hutter, F. Neural Architecture Search: A Survey. *J. Mach. Learn. Res.* **2019**, *20*, 1–21.
40. Real, E.; Moore, S.; Selle, A.; Saxena, S.; Suematsu, Y.L.; Tan, J.; Le, Q.V.; Kurakin, A. Large-Scale Evolution of Image Classifiers. In *Proceedings of the 34th International Conference on Machine Learning (ICML), Sydney, Australia, 6–11 August 2017*; PMLR (Proceedings of Machine Learning Research): Cambridge, MA, USA, 2017; pp. 2902–2911.
41. Real, E.; Aggarwal, A.; Huang, Y.; Le, Q.V. Regularized Evolution for Image Classifier Architecture Search. *Proc. AAAI Conf. Artif. Intell.* **2019**, *33*, 4780–4789. [CrossRef]
42. Jamieson, K.; Talwalkar, A. Non-Stochastic Best Arm Identification and Hyperparameter Optimization. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS), Cadiz, Spain, 9–11 May 2016*; PMLR (Proceedings of Machine Learning Research): Cambridge, MA, USA, 2016; pp. 240–248.
43. Li, L.; Jamieson, K.; DeSalvo, G.; Rostamizadeh, A.; Talwalkar, A. Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization. *J. Mach. Learn. Res.* **2018**, *18*, 1–52.
44. Dong, X.; Yang, Y. NAS-Bench-201: Extending the Scope of Reproducible Neural Architecture Search. In Proceedings of the 8th International Conference on Learning Representations (ICLR), Virtually, 26 April–1 May 2020.
45. Dhahir, N.; Rababa, M.; Koirala, B.; AbuGhazaleh, A.; AlSobeh, A. Rapid detection of Salmonella Typhimurium microcolonies using phase-contrast imaging and deep learning. *J. Food Prot.* **2026**, 100934. [CrossRef] [PubMed]
46. Jocher, G.; Qiu, J.; Liu, M.; Lyu, S.; Akyon, F.C.; Kalfaoglu, M.E. Ultralytics YOLO26: Unified Real-Time End-to-End Vision Models, 2026. *arXiv* **2026**, arXiv:2606.03748. [CrossRef]
47. Wang, A.; Liu, L.; Chen, H.; Lin, Z.; Han, J.; Ding, G. YOLOE: Real-Time Seeing Anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Honolulu, HI, USA, 19–23 October 2025*; IEEE: New York, NY, USA, 2025; pp. 24591–24602. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.